
Universal Intelligence: A Definition of Machine Intelligence

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Abstract

A fundamental problem in artificial intelligence is that nobody really knows what intelligence is. The problem is especially acute when we need to consider artificial systems which are significantly different to humans. In this paper we approach this problem in the following way: We take a number of well known informal definitions of human intelligence that have been given by experts, and extract their essential features. These are then mathematically formalised to produce a general measure of intelligence for arbitrary machines. We believe that this equation formally captures the concept of machine intelligence in the broadest reasonable sense. We then show how this formal definition is related to the theory of universal optimal learning agents. Finally, we survey the many other tests and definitions of intelligence that have been proposed for machines.

Keywords

AIXI; Complexity theory; Intelligence; Theoretical foundations; Turing test; Intelligence definitions; Intelligence tests.

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1 Introduction

“Innumerable tests are available for measuring intelligence, yet no one is quite certain of what intelligence is, or even just what it is that the available tests are measuring.”
 — R. L. Gregory [Gre98]

What is intelligence? It is a concept that we use in our daily lives that seems to have a fairly concrete, though perhaps naive, meaning. We say that our friend who got an A in his calculus test is very intelligent, or perhaps our cat who has learnt to go into hiding at the first mention of the word “vet”. Although this intuitive notion of intelligence presents us with no difficulties, if we attempt to dig deeper and define it in precise terms we find the concept to be very difficult to nail down. Perhaps the ability to learn quickly is central to intelligence? Or perhaps the total sum of one’s knowledge is more important? Perhaps communication and the ability to use language play a central role? What about “thinking” or the ability to perform abstract reasoning? How about the ability to be creative and solve problems? Intelligence involves a perplexing mixture of concepts, many of which are equally difficult to define.

Psychologists have been grappling with these issues ever since humans first became fascinated with the nature of the mind. Debates have raged back and forth concerning the correct definition of intelligence and how best to measure the intelligence of individuals. These debates have in many instances been very heated as what is at stake is not merely a scientific definition, but a fundamental issue of how we measure and value humans: Is one employee smarter than another? Are men on average more intelligent than women? Are white people smarter than black people? As a result intelligence tests, and their creators, have on occasion been the subject of intense public scrutiny. Simply determining whether a test, perhaps quite unintentionally, is partly a reflection of the race, gender, culture or social class of its creator is a subtle, complex and often politically charged issue [Gou81, HM96]. Not surprisingly, many have concluded that it is wise to stay well clear of this topic.

In reality the situation is not as bad as it is sometimes made out to be. Although the details of the definition are debated, in broad terms a fair degree of consensus about the scientific definition of intelligence and how to measure it has been achieved [Got97a, SB86]. Indeed it is widely recognised that when standard intelligence tests are correctly applied and interpreted, they all measure approximately the same thing [Got97a]. Furthermore, what they measure is both stable over time in individuals and has significant predictive power, in particular for future academic performance and other mentally demanding pursuits. The issues that continue to draw debate are the questions such as whether the tests test only a part or a particular type of intelligence, or whether they are somehow biased towards a particular group or set of mental skills. Great effort has gone into dealing with these issues, but they are difficult problems with no easy solutions.

Somewhat disconnected from this exists a parallel debate over the nature of intelligence in the context of machines. While the debate is less politically charged, in some ways the central issues are even more difficult. Machines can have physical forms, sensors, actuators, means of communication, information processing abilities and environments that are totally unlike those that we experience. This makes the concept of “machine intelligence” particularly difficult to get a handle on. In some cases, a machine may display properties that we equate with human intelligence, in such cases it might be

reasonable to describe the machine as also being intelligent. In other situations this view is far too limited and anthropocentric. Ideally we would like to be able to measure the intelligence of a wide range of systems; humans, dogs, flies, robots or even disembodied systems such as chat-bots, expert systems, classification systems and prediction algorithms [Joh92, Alb91].

One response to this problem might be to develop specific kinds of tests for specific kinds of entities; just as intelligence tests for children differ to intelligence tests for adults. While this works well when testing humans of different ages, it comes undone when we need to measure the intelligence of entities which are profoundly different to each other in terms of their cognitive capacities, speed, senses, environments in which they operate, and so on. To measure the intelligence of such diverse systems in a meaningful way we must step back from the specifics of particular systems and establish the underlying fundamentals of what it is that we are really trying to measure.

The difficulty of developing an abstract and highly general notion of intelligence is readily apparent. Consider, for example, the memory and numerical computation tasks that appear in some intelligence tests and which were once regarded as defining hallmarks of human intelligence. We now know that these tasks are absolutely trivial for a machine and thus do not appear to test the machine's intelligence in any meaningful sense. Indeed even the mentally demanding task of playing chess can be largely reduced to brute force search [HCH95]. What else may in time be possible with relatively simple algorithms running on powerful machines is hard to say. What we can be sure of is that as technology advances, our concept of intelligence will continue to evolve with it.

How then are we to develop a concept of intelligence that is applicable to all kinds of systems? Any proposed definition must encompass the essence of human intelligence, as well as other possibilities, in a consistent way. It should not be limited to any particular set of senses, environments or goals, nor should it be limited to any specific kind of hardware, such as silicon or biological neurons. It should be based on principles which are fundamental and thus unlikely to alter over time. Furthermore, the definition of intelligence should ideally be formally expressed, objective, and practically realisable as an effective intelligence test.

In this paper we approach the problem of defining machine intelligence as follows:

Section 2 overviews well known theories, definitions and tests of intelligence that have been developed by psychologists. Our objective in this section is to gain an understanding of the essence of intelligence in the broadest possible terms. In particular we are interested in commonly expressed ideas that could be applied to arbitrary systems and contexts, not just humans.

Section 3 takes these key ideas and formalises them. This leads to *universal intelligence*, our proposed formal definition of machine intelligence. We then examine some of the properties of universal intelligence, such as its ability to sensibly order simple learning algorithms and connections to the theory of universal optimal learning agents.

Section 4 overviews other definitions and tests of machine intelligence that have been proposed. Although surveys of the Turing test and its many variants exist, for example [SCA00], as far as we know this section is the first general survey of definitions and tests of machine intelligence. Given how fundamental this is to the field of artificial intelligence, the absence of such a survey is quite remarkable. For any field to mature as a science, questions of definition and measurement must be meticulously investigated. We conclude

our survey with a summary comparison of the various proposed tests and definitions of machine intelligence.

Section 5 ends the paper with discussion, responses to criticisms, conclusions and directions for future research.

The genesis of this work lies in Hutter’s universal optimal learning agent, AIXI, described in 2, 12, 60 and 300 pages in [Hut01b, Hut01a, Hut05, Hut07b], respectively. In this work, an order relation for intelligent agents is presented, with respect to which the provably optimal AIXI agent is maximal. The universal intelligence measure presented here is a derivative of this order relation. A short description of the universal intelligence measure appeared in [LH05], from which two articles followed in the popular scientific press [GR05, Fié05]. An 8 page paper on universal intelligence appeared in [LH06b], followed by an updated poster presentation [LH06a]. In the current paper we explore universal intelligence in much greater detail, in particular the way in which it relates to mainstream views on human intelligence and other proposed definitions of machine intelligence.

2 Natural Intelligence

Human intelligence is an enormously rich topic with a complex intellectual, social and political history. For an overview the interested reader might want to consult “Handbook of Intelligence” [Ste00] edited by R. J. Sternberg. Our objective in this section is simply to sketch a range of tests, theories and definitions of human and animal intelligence. We are particularly interested in common themes and general perspectives on intelligence that could be applicable to many kinds of systems, as these will form the foundation of our definition of machine intelligence in the next section.

2.1 Human intelligence tests

Contrary to popular public opinion, most psychologists believe that the usual tests of intelligence, such as IQ tests, reliably measure something important in humans [NBB⁺96, Got97b]. In fact, standard intelligence tests are among the most statistically stable and reliable of psychological tests. Furthermore, it is well known that these scores are a good predictor of various things, such as academic performance. The question then is not whether these tests are useful or measure something meaningful, but rather whether what they measure is indeed “intelligence”. Some experts believe that they do, while others think that they only succeed in measuring certain aspects of, or types of, intelligence.

The first modern style intelligence test was developed by the French psychologist Alfred Binet in 1905. Binet believed that intelligence was best studied by looking at relatively complex mental tasks, unlike earlier tests developed by Francis Galton which focused on reaction times, auditory discrimination ability, physical coordination and so on. Binet’s test consisted of 30 short tasks related to everyday problems such as; naming parts of the body, comparing lengths and weights, counting coins, remembering digits and definitions of words. For each task category there were a number of problems of increasing difficulty. The child’s results were obtained by normalising their raw score against peers of the same age. Initially his test was designed to measure the mental performance of children with learning problems [BS05]. Later versions were also developed for normal

children [Bin11]. It was found that Binet’s test results were a good predictor of children’s academic performance.

Lewis Terman of Stanford University developed an English version of Binet’s test. As the age norms for French children did not correspond well with American children, he revised Binet’s test in various ways, in particular he increased the upper age limit. This resulted in the now famous Stanford-Binet test [TM50]. This test formed the basis of a number of other intelligence tests, such as the Army Alpha and Army Beta tests which were used to classify recruits. Since its development, the Stanford-Binet has been periodically revised, with updated versions being widely used today.

David Wechsler believed that the original Binet tests were too focused on verbal skills and thus disadvantaged certain otherwise intelligent individuals, for example the deaf or people who did not speak the test language as a first language. To address this problem, he proposed that tests should contain a combination of both verbal and nonverbal problems. He also believed that in addition to an overall IQ score, a profile should be produced showing the performance of the individual in the various areas tested. Borrowing significantly from the Stanford-Binet, the US army Alpha test, and others, he developed a range of tests targeting specific age groups from preschoolers up to adults [Wec58]. Due in part to problems with revisions of the Stanford-Binet test in the 1960’s and 1970’s, Wechsler’s tests became the standard. They continue to be well respected and widely used.

Owing to a common lineage, modern versions of the Wechsler and the Stanford-Binet have a similar basic structure [Kau00]. Both test the individual in a number of verbal and non-verbal ways. In the case of a Stanford-Binet the test is broken up into 5 key areas: Fluid reasoning, knowledge, quantitative reasoning, visual-spatial processing, and working memory. In the case of the Wechsler Adult Intelligence Scale (WAIS-III), the verbal tests include areas such as knowledge, basic arithmetic, comprehension, vocabulary, and short term memory. Non-verbal tests include picture completion, spatial perception, problem solving, symbol search and object assembly.

As part of an effort to make intelligence tests more culture neutral John Raven developed the progressive matrices test [Rav00]. In this test each problem consists of a short sequence of basic shapes. For example, a circle in a box, then a circle with a cross in the middle followed by a circle with a triangle inside. The test subject then has to select from a second list the image that best continues the pattern. Simple problems have simple patterns, while difficult problems have more subtle and complex patterns. In each case however, the simplest pattern that can explain the observed sequence is the one that correctly predicts its continuation. Thus, not only is the ability to recognise patterns tested, but also the ability to evaluate the complexity of different explanations and then correctly apply the philosophical principle of Occam’s razor. We will return to Occam’s razor and its importance in intelligence testing in Subsection 3.3 when considering machine intelligence.

Today several different versions of the Raven test exist designed for different age groups and ability levels. As the tests depend strongly on the ability to identify abstract patterns, rather than knowledge, they are considered to be some of the most “g-loaded” intelligence tests available (see Subsection 2.5). The Raven tests remain in common use today, particularly when it is thought that culture or language bias could be an issue.

The intelligence quotient, or IQ, was originally introduced by Stern [Ste12]. It was computed by taking the age of a child as estimated by their performance in the intelligence

test, and then dividing this by their true biological age and multiplying by 100. Thus a 10 year old child whose mental performance was equal to that of a normal 12 year old, had an IQ of 120. As the concept of mental age has now been discredited, and was never applicable to adults anyway, modern IQ scores are simply normalised to a Gaussian distribution with a mean of 100. The standard deviation used varies: in the United States 15 is commonly used, while in Europe 25 is common. For children the normalising Gaussian is based on peers of the same age.

Whatever normalising distribution is used, by definition an individual's IQ is always an indication of their cognitive performance relative to some larger group. Clearly this would be problematic in the context of machines where the performance of some machines could be many orders of magnitude greater than others. Furthermore, the distribution of machine performance would be continually changing due to advancing technology. Thus, for our purposes, an absolute measure will be more meaningful than a traditional IQ type of measure.

For an overview of the history of intelligence testing and the structure of modern tests, see [Kau00].

2.2 **Animal intelligence tests**

Testing the intelligence of animals is of particular interest to us as it moves beyond strictly human focused concepts of intelligence and testing methods. Difficult problems in human intelligence testing, such as bias due to language differences or physical handicap, become even more difficult if we try to compare animals with different perceptual and cognitive capacities. Even within a single species measurement is difficult as it is not always obvious how to conduct the tests, or even what should be tested for. Furthermore, as humans devise the tests, there is a persistent danger that the tests may be biased in terms of our sensory, motor, and motivational systems [Mac85]. For example, it is known that rats can learn some types of relationships much more easily through smell rather than other senses [SK74]. Furthermore, while an IQ test for children might in some sense be validated by its ability to predict future academic or other success, it is not always clear how to validate an intelligence test for animals. If survival or the total number of offspring was a measure of success, then bacteria might be the most intelligent life on earth!

As is often the case when we try to generalise concepts, abstraction is necessary. When attempting to measure the intelligence of lower animals it is necessary to focus on simple things like short and long term memory, the forming of associations, the ability to generalise simple patterns and make predictions, simple counting and basic communication. It is only with relatively intelligent social animals, such as birds and apes, that more sophisticated properties such as deception, imitation and the ability to recognise oneself become important. For simpler animals, the focus is more on the animal's essential information processing capacity. For example, the work on understanding the capacity of ants to remember patterns when retracing a path back to a source of food without the aid of pheromones [RR86].

One interesting difficulty when testing animal intelligence is that we are unable to directly explain to the animal what its goal is. Instead, we have to guide the animal towards a problem by carefully rewarding selected behaviours with something like food. In general, when testing machine intelligence we face a similar problem in that we cannot assume that a machine will have a sufficient level of language comprehension to be able to under-

stand commands. Thus a simple solution is to use basic “rewards” to guide behaviour, as we do with animals. Although this approach is extremely general, one difficulty is that solving the task, and simply learning what the task is, become confounded and thus the results need to be interpreted carefully [Zen97]. Due to our need for generality, we will use this reward based approach for our formal measure of machine intelligence. Specifically, we will adopt the reinforcement learning framework from artificial intelligence (see Subsection 3.1).

For good overviews of animal intelligence research see [Zen00] or [HP94].

2.3 Desirable properties of an intelligence test

There are many properties that a good test of human intelligence should have. One important property is that the test should be repeatable, in the sense that it consistently returns about the same score for a given individual. For example, the test subject should not be able to significantly improve their performance if tested again a short time later. Statistical variability can also be a problem in short tests. Longer tests help in this regard, however they are naturally more costly to administer.

Another important reliability factor is the bias that might be introduced by the individual administering the test. Purely written tests avoid this problem as there is minimal interaction between the tested individual and the tester. However this lack of interaction also has disadvantages as it may mean that other sources of bias, such as cultural differences, language problems or even something as simple as poor eyesight, might not be properly identified. Thus, even in a written test the individual being tested should first be examined by an expert in order to ensure that the test is appropriate.

Cultural bias in particular is a difficult problem, and tests should be designed to minimise this problem where possible, or at least detect potential bias problems when they occur. One way to do this is to test each ability in multiple ways, for example both verbally and visually. While language is an obvious potential source of cultural bias, more subtle forms of bias are difficult to detect and remedy. For example, different cultures emphasise different cognitive abilities, and thus it is difficult, perhaps impossible, to compare intelligence scores in a way that is truly objective. In part this is a question of what intelligence is. Indeed the problem of how to weight performance in different areas is fundamental and we will need to face it again in the context of our formal definition of machine intelligence.

When testing large numbers of individuals, for example when testing army recruits, the cost of administering the test becomes important. In these cases less accurate but more economical test procedures may be used, such as purely written tests without any direct interaction between the individuals being tested and a psychologist.

An intelligence test should be valid in the sense that it appears to be testing what it claims it is testing for. One way to check this is to show that the test produces results consistent with other manifestations of intelligence. A test should also have predictive power, for example the ability to predict future academic performance. This ensures that what is being measured is somehow meaningful, beyond just the ability to answer the questions in the test.

Standard intelligence tests such as a modern Stanford-Binet are thoroughly tested for years on the above criteria, and many others, before they are ready for wide spread use. Many of these desirable properties, such as reliability, tester bias, cost and validity, are also

relevant to tests of machine intelligence. To some extent they are also relevant to formal definitions of intelligence. We will return to these desirable properties when analysing our definition of machine intelligence in Subsection 3.5, and later when comparing tests of machine intelligence in Subsection 4.3.

2.4 Static vs. dynamic tests

Stanford-Binet, Wechsler, Raven progressive matrices, and indeed most standard intelligence tests, are all examples of “static tests”. By this we mean that they test an individual’s knowledge and ability to solve one-off problems. They do not directly measure the ability to learn and adapt over time. If an individual was good at learning and adapting then we might expect this to be reflected in their total knowledge and thus picked up in a static test. However, it could be that an individual has a great capacity to learn, but that this is not reflected in their knowledge due to limited education. In which case, if we consider the capacity to learn and adapt rather than the sum of knowledge to be a defining characteristic of intelligence, then to class an individual as unintelligent due to limited access to education would be a mistake.

What is needed is a more direct test of an individual’s ability to learn and adapt: A so called “dynamic test” [SG02] (for related work see also [JLW77]). In a dynamic test the test subject interacts over a period of time with the tester, who now becomes a kind of teacher. The tester’s task is to present the individual with a series of problems. After each attempt at solving a problem, the tester provides feedback to the individual who then has to adapt their behaviour accordingly in order to solve the next problem.

Although dynamic tests could in theory be very powerful, they are not yet well established due to a number of difficulties. One of the drawbacks is that they require a much greater degree of interaction between the test subject and the tester. This makes dynamic testing more costly to perform and increases the danger of tester bias.

Dynamic testing is of particular interest to us because in a formal test for machines it appears that we can overcome these problems by automating the role of the tester.

2.5 Theories of human intelligence

Complementary to the experimental study of human intelligence, theories have been developed that attempt to better characterise the fundamental nature of intelligence. It is useful for us to briefly sketch this work as some of these issues have parallels within the context of machine intelligence.

One central question is whether intelligence should be viewed as one ability, or many. On one side of the debate are the theories that view intelligence as consisting of many different components and that identifying these components is important to understanding intelligence. Different theories propose different ways to do this. One of the first was Thurstone’s “multiple-factors” theory which considers seven “primary mental abilities”: verbal comprehension, word fluency, number facility, spatial visualisation, associative memory, perceptual speed and reasoning [Thu38]. Another approach is Sternberg’s “Triarchic Mind” which breaks intelligence down into analytical intelligence, creative intelligence, and practical intelligence [Ste85], however this model is now considered outdated, even by Sternberg himself.

Taking the number of components to an extreme is Guilford’s “Structure of Intellect” theory. Under this theory there are three fundamental dimensions: Contents, operations, and products. Together these give rise to 120 different categories [Gui67] (in later work this increased to 150 categories). This theory has been criticised due to the fact that measuring such precise combinations of cognitive capacities in individuals seems to be infeasible and thus it is difficult to experimentally study such a fine grained model of intelligence.

A recently popular approach is Gardner’s “multiple intelligences” where he argues that the components of human intelligence are sufficiently separate that they are actually different “intelligences” [Gar93]. Based on the structure of the human brain he identifies these intelligences to be linguistic, musical, logical-mathematical, spatial, bodily kinaesthetic, intra-personal and inter-personal intelligence. Although Gardner’s theory of multiple intelligences has certainly captured the imagination of the public, it remains to be seen to what degree it will have a lasting impact in professional circles.

At the other end of the spectrum is the work of Spearman and those that have followed in his footsteps. Here intelligence is seen as a very general mental ability that underlies and contributes to all other mental abilities. As evidence they point to the fact that an individual’s performance levels in reasoning, association, linguistic, spatial thinking, pattern identification etc. are positively correlated. Spearman called this positive statistical correlation between different mental abilities the “ g -factor”, where g stands for “general intelligence” [Spe27]. Because standard IQ tests measure a range of key cognitive abilities, from a collection of scores on different cognitive tasks we can estimate an individual’s g -factor. Some who consider the generality of intelligence to be of primary importance take the g -factor to be the very definition of intelligence [Got02].

A well known refinement to the g -factor theory due to Cattell is to distinguish between, “fluid intelligence”, which is a very general and flexible innate ability to deal with problems and complexity, and “crystallized intelligence”, which measures the knowledge and abilities that an individual has acquired over time [Cat87]. For example, while an adolescent may have a similar level of fluid intelligence to that of an adult, their level of crystallized intelligence is typically lower due to less life experience [Hor70]. Although it is difficult to determine to what extent these two influence each other, the distinction is an important one because it captures two distinct notions of what the word “intelligence” means.

As the g -factor is simply the statistical correlation between different kinds of mental abilities, it is not fundamentally inconsistent with the view that intelligence can have multiple aspects or dimensions. Thus a synthesis of the two perspectives is possible by viewing intelligence as a hierarchy with the g -factor at its apex and increasing levels of specialisation for the different aspects of intelligence forming branches [Car93]. For example, an individual might have a high g -factor, which contributes to all of their cognitive abilities, but also have an especially well developed musical sense. This hierarchical view of intelligence is now quite popular [NBB⁺96].

2.6 Ten definitions of human intelligence

“Viewed narrowly, there seem to be almost as many definitions of intelligence as there were experts asked to define it.” — R. J. Sternberg quoted in [Gre98]

In this subsection and the next we will overview a range of definitions of intelligence

that have been given by psychologists. Many of these definitions are well known. Although the definitions differ, there are reoccurring features; in some cases these are explicitly stated, while in others they are more implicit. We start by considering ten definitions that take, in our view, a similar perspective:

“It seems to us that in intelligence there is a fundamental faculty, the alteration or the lack of which, is of the utmost importance for practical life. This faculty is judgement, otherwise called good sense, practical sense, initiative, the faculty of adapting oneself to circumstances.” A. Binet [BS05]

“The capacity to learn or to profit by experience.” W. F. Dearborn quoted in [Ste00]

“Ability to adapt oneself adequately to relatively new situations in life.” R. Pinter quoted in [Ste00]

“A person possesses intelligence insofar as he has learned, or can learn, to adjust himself to his environment.” S. S. Colvin quoted in [Ste00]

“We shall use the term ‘intelligence’ to mean the ability of an organism to solve new problems ...” W. V. Bingham [Bin37]

“A global concept that involves an individual’s ability to act purposefully, think rationally, and deal effectively with the environment.” D. Wechsler [Wec58]

“Individuals differ from one another in their ability to understand complex ideas, to adapt effectively to the environment, to learn from experience, to engage in various forms of reasoning, to overcome obstacles by taking thought.” American Psychological Association [NBB⁺96]

“...I prefer to refer to it as ‘successful intelligence.’ And the reason is that the emphasis is on the use of your intelligence to achieve success in your life. So I define it as your skill in achieving whatever it is you want to attain in your life within your sociocultural context — meaning that people have different goals for themselves, and for some it’s to get very good grades in school and to do well on tests, and for others it might be to become a very good basketball player or actress or musician.” R. J. Sternberg [Ste03]

“Intelligence is part of the internal environment that shows through at the interface between person and external environment as a function of cognitive task demands.” R. E. Snow quoted in [Sla01]

“...certain set of cognitive capacities that enable an individual to adapt and thrive in any given environment they find themselves in, and those cognitive capacities include things like memory and retrieval, and problem solving and so forth. There’s a cluster of cognitive abilities that lead to successful adaptation to a wide range of environments.” D. K. Simonton [Sim03]

Perhaps the most elementary common feature of these definitions is that intelligence is seen as a property of an individual who is interacting with an external environment, problem or situation. Indeed, at least this much is common to practically all proposed definitions of intelligence.

Another common feature is that an individual's intelligence is related to their ability to succeed or "profit". The notion of success or profit implies the existence of some kind of objective or goal. What the goal is, is not specified, indeed individuals' goals may be varied. The important thing is that the individual is able to carefully choose their actions in a way that leads to them accomplishing their goals. The greater this capacity to succeed with respect to various goals, the greater the individual's intelligence.

The strong emphasis on learning, adaption and experience in these definitions implies that the environment is not fully known to the individual and may contain new situations that could not have been anticipated in advance. Thus intelligence is not the ability to deal with a fully known environment, but rather the ability to deal with some range of possibilities which cannot be wholly anticipated. What is important then is that the individual is able to quickly learn and adapt so as to perform as well as possible over a wide range of environments, situations, tasks and problems. Collectively we will refer to these as "environments", similar to some of the definitions above.

Bringing these key features together gives us what we believe to be the essence of intelligence in its most general form:

Intelligence measures an agent's ability to achieve goals in a wide range of environments.

We take this to be our informal working definition of intelligence. In the next section we will use this definition as the starting point from which we will construct a formal definition of machine intelligence. However before we proceed further, the reader may wish to revise the 10 definitions above to ensure that the definition we have adopted is indeed reasonable.

2.7 More definitions of human intelligence

Of course many other definitions of intelligence have been proposed over the years. Usually they are not entirely incompatible with our informal definition, but rather stress different aspects of intelligence. In this subsection we will survey some of these other definitions and compare them to the position we have taken. For an even more extensive collection of definitions of intelligence, indeed the largest collection that we are aware of, visit our online collection [LH07].

The following is an especially interesting definition as it was given as part of a group statement signed by 52 experts in the field. As such it obviously represents a fairly mainstream perspective:

"Intelligence is a very general mental capability that, among other things, involves the ability to reason, plan, solve problems, think abstractly, comprehend complex ideas, learn quickly and learn from experience." [Got97a]

Reasoning, planning, solving problems, abstract thinking, learning from experience and so on, these are all mental abilities that allow us to successfully achieve goals. If we were missing any one of these capacities, we would clearly be less able to successfully deal with such a wide range of environments. Thus, these capacities are implicit in our definition also. The difference is that our definition does not attempt to specify what capabilities might be needed, something which is clearly very difficult and would depend

on the particular tasks that the agent must deal with. Our approach is to consider intelligence to be the *effect* of capacities such as those listed above. It is not the result of having any specific set of capacities. Indeed, intelligence could also be the effect of many other capacities, some of which humans may not have. In summary, our definition is not in conflict with the above definition, rather it is that our definition is more abstract and general.

“... in its lowest terms intelligence is present where the individual animal, or human being, is aware, however dimly, of the relevance of his behaviour to an objective. Many definitions of what is indefinable have been attempted by psychologists, of which the least unsatisfactory are 1. the capacity to meet novel situations, or to learn to do so, by new adaptive responses and 2. the ability to perform tests or tasks, involving the grasping of relationships, the degree of intelligence being proportional to the complexity, or the abstractness, or both, of the relationship.”
J. Drever [Dre52]

This definition has many similarities to ours. Firstly, it emphasises the agent’s ability to choose its actions so as to achieve an objective, or in our terminology, a goal. It then goes on to stress the agent’s ability to deal with situations which have not been encountered before. In our terminology, this is the ability to deal with a wide range of environments. Finally, this definition highlights the agent’s ability to perform tests or tasks, something which is entirely consistent with our performance orientated perspective of intelligence.

“Intelligence is not a single, unitary ability, but rather a composite of several functions. The term denotes that combination of abilities required for survival and advancement within a particular culture.” A. Anastasi [Ana92]

This definition does not specify exactly which capacities are important, only that they should enable the individual to survive and advance with the culture. As such this is a more abstract “success” orientated definition of intelligence, like ours. Naturally, culture is a part of the agent’s environment.

“The ability to carry on abstract thinking.” L. M. Terman quoted in [Ste00]

This is not really much of a definition as it simply shifts the problem of defining intelligence to the problem of defining abstract thinking. The same is true of many other definitions that refer to things such as imagination, creativity or consciousness. The following definition has a similar problem:

“The capacity for knowledge, and knowledge possessed.” V. A. C. Henmon [Hen21]

What exactly constitutes “knowledge”, as opposed to perhaps data or information? For example, does a library contain a lot of knowledge, and if so, is it intelligent? Or perhaps the internet? Modern concepts of the word knowledge stress the fact that the information has to be in some sense properly contextualised so that it has meaning. Defining this more precisely appears to be difficult however. Because this definition of intelligence dates from 1921, perhaps it reflects pre-information age thinking when computers with vast storage capacities did not exist.

Nonetheless, our definition of intelligence is not entirely inconsistent with the above definition in that an individual may be required to know many things, or have a significant capacity for knowledge, in order to perform well in some environments. However our definition is broader in that knowledge, or the capacity for knowledge, is not by itself sufficient. We require that the knowledge can be used effectively for some purpose. Indeed unless information can be effectively utilised for a number of purposes, it seems reasonable to consider it to be merely “data”, rather than “knowledge”.

“The capacity to acquire capacity.” H. Woodrow quoted in [Ste00]

The definition of Woodrow is typical of those which emphasise not the current ability of the individual, but rather the individual’s ability to expand and develop new abilities. This is a fundamental point of divergence for many views on intelligence. Consider the following question: Is a young child as intelligent as an adult? From one perspective, children are very intelligent because they can learn and adapt to new situations quickly. On the other hand, the child is unable to do many things due to a lack of knowledge and experience and thus will make mistakes an adult would know to avoid. These need not just be physical acts, they could also be more subtle things like errors in reasoning as their mind, while very malleable, has not yet matured. In which case, perhaps their intelligence is currently low, but will increase with time and experience?

Fundamentally, this difference in perspective is a question of time scale: Must an agent be able to tackle some task immediately, or perhaps after a short period of time during which learning can take place, or perhaps it only matters that they can eventually learn to deal with the problem? Being able to deal with a difficult problem immediately is a matter of experience, rather than intelligence. While being able to deal with it in the very long run might not require much intelligence at all, for example, simply trying a vast number of possible solutions might eventually produce the desired results. Intelligence then seems to be the ability to adapt and learn as quickly as possible given the constraints imposed by the problem at hand. It is this insight that we will use to neatly deal with temporal preference when defining machine intelligence (see *Measure of success* in Subsection 3.2).

“Intelligence is a general factor that runs through all types of performance.”
A. Jensen

At first this might not look like a definition of intelligence, but it makes an important point: Intelligence is not really the ability to do anything in particular, rather it is a very general ability that affects many kinds of performance. Conversely, by measuring many different kinds of performance we can estimate an individual’s intelligence. This is consistent with our definition’s emphasis on the agent’s generality.

“Intelligence is what is measured by intelligence tests.” E. Boring [Bor23]

Boring’s famous definition of intelligence takes this idea a step further. If intelligence is not the ability to do anything in particular, but rather an abstract ability that indirectly affects performance in many tasks, then perhaps it is most concretely described as the ability to do the kind of abstract problems that appear in intelligence tests? In which case, Boring’s definition is not as facetious as it first appears.

This definition also highlights the fact that the concept of intelligence, and how it is measured, are intimately related. In the context of this paper we refer to these as definitions of intelligence, and tests of intelligence, respectively.

3 A Definition of Machine Intelligence

“Indeed the guiding inspiration of cognitive science is that at a suitable level of abstraction, a theory of natural intelligence should have the same basic form as the theories that explain sophisticated computer systems.”

— J. Haugeland [Hau81]

Having presented a very general informal definition of intelligence in Subsection 2.6, we will now proceed to formalise this definition mathematically in a way that is appropriate for machines. We will then study some of the properties of this definition in the remainder of this section.

3.1 Basic agent-environment framework

Consider again our informal definition of intelligence:

Intelligence measures an agent’s ability to achieve goals in a wide range of environments.

This definition contains three essential components: An agent, environments and goals. Clearly, the agent and the environment must be able to interact with each other, specifically, the agent needs to be able to send signals to the environment and also receive signals being sent from the environment. Similarly, the environment must be able to receive and send signals to the agent. In our terminology we will adopt the agent’s perspective on these communications and refer to the signals sent from the agent to the environment as *actions*, and the signals sent from the environment to the agent as *perceptions*.

Our definition of an agent’s intelligence also requires there to be some kind of goal for the agent to try to achieve. Perhaps an agent could be intelligent, in an abstract sense, without having any objective to apply its intelligence to. Or perhaps the agent has no desire to exercise its intelligence in a way that effects its environment. In either case, the agent’s intelligence would be unobservable and, more importantly, of no practical consequence. Intelligence then, at least the concrete kind that interests us, comes into effect when the agent has an objective that it actively pursues by interacting with its environment. Here we will refer to this objective as its *goal*.

The existence of a goal raises the problem of how the agent knows what the goal is. One possibility would be for the goal to be known in advance and for this knowledge to be built into the agent. The problem with this however is that it limits each agent to just one goal. We need to allow agents that are more flexible, specifically, we need to be able to inform the agent of what the goal is. For humans this is easily done using language. In general however, the possession of a sufficiently high level of language is too strong an assumption to make about the agent. Indeed, even for something as intelligent as a dog or a cat, direct explanation is not very effective.

Fortunately there is another possibility which is, in some sense, a blend of the above two. We define an additional communication channel with the simplest possible semantics: A signal that indicates how good the agent’s current situation is. We will call this signal the *reward*. The agent’s goal is then simply to maximise the amount of reward it receives. So in a sense its goal is fixed. This is not particularly limiting however, as we have not said anything about what causes different levels of reward to occur. In a complex setting

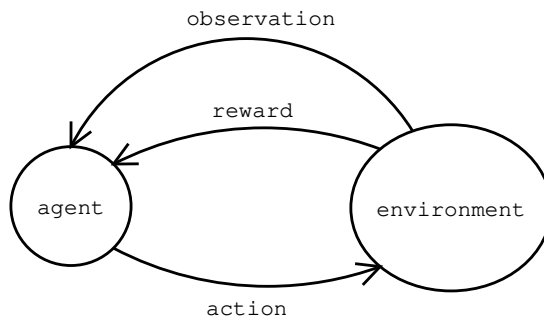


Figure 1: The agent and the environment interact by sending action, observation and reward signals to each other.

the agent might be rewarded for winning a game or solving a puzzle. If the agent is to succeed in its environment, that is, receive a lot of reward, it must learn about the structure of the environment and in particular what it needs to do in order to get reward. Thus from a broad perspective, the goal is flexible. Not surprisingly, this is exactly the way in which we condition an animal to achieve a goal: By selectively rewarding certain behaviours (see Subsection 2.2). In a narrow sense the animal’s goal is fixed, perhaps to get more treats to eat, but in a broader sense it is flexible as it may require doing a trick or solving a puzzle of our choosing.

In our framework we will include the reward signal as a part of the perception generated by the environment. The perceptions also contain a non-reward part, which we will refer to as *observations*. This now gives us the complete system of interacting agent and environment, as illustrated in Figure 1. The goal, in the broad flexible sense, is implicitly defined by the environment as this is what defines when rewards are generated. Thus, in the framework as we have defined it, to test an agent in any given way it is sufficient to fully define the environment.

This widely used and very flexible structure is in itself nothing new. In artificial intelligence it is the framework used in reinforcement learning [SB98]. By appropriately renaming things, it also describes the controller-plant framework used in control theory [BT96]. The interesting point for us is that this setup follows naturally from our informal definition of intelligence and our desire to keep things as general as possible. The only difficulty was how to deal with the notion of success, or profit. This required the existence of some kind of an objective or goal. The most flexible and elegant way to bring this into the framework was to use a simple reward signal.

3.1 Example. To make this model more concrete, consider the following “Two Coins Game”. In each cycle two 50¢ coins are tossed. Before the coins settle the player must guess at the number of heads that will result: either 0, 1, or 2. If the guess is correct the player gets to keep both coins and then two new coins are produced and the game repeats. If the guess is incorrect the player does not receive any coins, and the game is repeated.

In terms of the agent-environment model, the player is the agent and the system that produces all the coins, tosses them and distributes the reward when appropriate, is the environment. The agent’s actions are its guesses at the number of heads in each iteration of the game: 0, 1 or 2. The observation is the state of the coins when they settle, and the reward is either \$0 or \$1.

It is easy to see that for unbiased coins the most likely outcome is 1 head and thus the optimal strategy for the agent is to always guess 1. However if the coins are significantly biased it might be optimal to guess either 0 or 2 heads depending on the bias. If this were the case, then after a number of iterations of the game an intelligent agent would realise that the coins were probably biased and change its strategy accordingly.

With a little imagination, seemingly any sort of game, challenge, problem or test can be expressed in this simple framework without too much effort. It should also be emphasised that this agent-environment framework says nothing about how the agent or the environment actually work; it only describes their roles.

3.2 Formal agent-environment framework

Having introduced the agent-environment framework, we will now formalise it, along with the other components of our informal definition of intelligence. We begin with agent-environment interaction.

Agent-environment interaction. The agent sends information to the environment by sending *symbols* from some finite set, for example, $\mathcal{A} := \{\text{left}, \text{right}, \text{up}, \text{down}\}$. We will call this set the *action space* and denote it by $\mathcal{A}$. Similarly, the environment sends signals to the agent with symbols from a finite set called the *perception space*, which we will denote $\mathcal{P}$. The *reward space*, denoted by $\mathcal{R}$, will always be a subset of the rational unit interval $[0, 1] \cap \mathbb{Q}$. Every perception consists of two separate parts; an observation and a reward. For example, we might have $\mathcal{P} := \{(\text{cold}, 0.0), (\text{warm}, 1.0), (\text{hot}, 0.3)\}$ where the first part describes what the agent observes (cold, warm or hot) and the second part describes the reward (0.0, 1.0, or 0.3).

To denote symbols being sent we will use the lower case variable names a , o and r for actions, observations and rewards respectively. We will also index these in the order in which they occur, thus a_1 is the agent's first action, a_2 is the second action and so on. The agent and the environment will take turns at sending symbols, starting with the environment. This produces a history of observations, rewards and actions which we will denote by, $o_1 r_1 a_1 o_2 r_2 a_2 o_3 r_3 a_3 o_4 \dots$. This turn taking behaviour is not a serious restriction, nor is the fact that the first signal sent is a perception.

The agent. Formally, the agent is a function, denoted by π , which takes the current history as input and chooses the next action as output. We do not want to restrict the agent in any way, in particular we do not require that it is deterministic. A convenient way of representing the agent then is as a probability measure over actions conditioned on the complete interaction history. Thus, $\pi(a_3 | o_1 r_1 a_1 o_2 r_2)$ is the probability of action a_3 in the third cycle, given that the current history is $o_1 r_1 a_1 o_2 r_2$. A deterministic agent is simply one that always assigns a probability of 1 to a single action for any given history. As the history that the agent can use to select its action expands indefinitely, the agent need not be Markovian. Indeed, how the agent produces its distribution over actions for any given history is left open. In artificial intelligence the agent will of course be a machine and so π will be a computable function. In general however, π could be anything: An algorithm that generates the digits of $\sqrt{e}$ as outputs, an uncomputable function, or even a human pushing buttons on a keyboard.

The environment. We define the environment, denoted by μ , in a similar way. Specifically, for any $k \in \mathbb{N}$ the probability of $o_k r_k$, given the current interaction history $o_1 r_1 a_1 o_2 r_2 a_2 \dots o_{k-1} r_{k-1} a_{k-1}$, is given by the probability measure $\mu(o_k r_k | o_1 r_1 a_1 o_2 r_2 a_2 \dots o_{k-1} r_{k-1} a_{k-1})$. For the moment we will not place any further restrictions on the environment.

3.2 Example. To illustrate this formalism, consider again the Two Coins Game introduced in Example 3.1. Let $\mathcal{P} := \{0, 1, 2\} \times \{0, 1\}$ be the perception space representing the number of heads after tossing the two coins and the value of the received reward. Likewise let $\mathcal{A} := \{0, 1, 2\}$ be the action space representing the agent's guess at the number of heads that will occur. Assuming two fair coins, we can represent this environment by μ :

$$\mu(o_k r_k | o_1 \dots a_{k-1}) := \begin{cases} \frac{1}{4} & \text{if } o_k = a_{k-1} \in \{0, 2\} \wedge r_k = 1, \\ \frac{3}{4} & \text{if } o_k \neq a_{k-1} \in \{0, 2\} \wedge r_k = 0, \\ \frac{1}{2} & \text{if } o_k = a_{k-1} = 1 \wedge r_k = 1, \\ \frac{1}{2} & \text{if } o_k \neq a_{k-1} = 1 \wedge r_k = 0, \\ 0 & \text{otherwise.} \end{cases}$$

An agent that performs well in this environment would be,

$$\pi(a_k | o_1 r_1 a_1 \dots o_k r_k) := \begin{cases} 1 & \text{for } a_k = 1, \\ 0 & \text{otherwise.} \end{cases}$$

That is, always guess that one head will be the result of the two coins being tossed. A more complex agent might keep count of how many heads occur in each cycle and then adapt its strategy if it seems that the coins are sufficiently biased.

Measure of success. Our next task is to formalise the idea of “profit” or “success” for an agent. Informally, we know that the agent must try to maximise the amount of reward it receives, however this could mean several different things. For example, one agent might quickly find a way to get a reward of 0.9 in every cycle. After 100 cycles it will have received a total reward of about 90 with an average reward per cycle of close to 0.9. A second agent might spend the first 80 cycles exploring different actions and their consequences, during which time its average reward might only be 0.2. Having done this exploration however, it might then know a way to get a reward of 1.0 in every cycle. Thus after 100 cycles its total reward is only $80 \times 0.2 + 20.0 = 36$, giving an average reward per cycle of just 0.36. After 1,000 cycles however, the second agent will be performing much better than the first.

Which agent is the better one? The answer depends on how we value reward in the near future versus reward in the more distant future. In some situations we may want our agent to perform well fairly quickly, in others we might only care that it eventually reaches a level of performance that is as high as possible.

A standard way of formalising this is to scale the value of rewards so that they decay geometrically into the future at a rate given by a discount parameter $\gamma \in (0, 1)$. For example, with $\gamma = 0.95$ a reward of 0.7 that is 10 time steps into the future would be given a value of $0.7 \times (0.95)^{10} \approx 0.42$. At 100 time steps into the future a reward of 0.7 would have a value of just over 0.004. By increasing γ towards 1 we weight long term

rewards more heavily, conversely by reducing it we weight them less so. In other words, this parameter controls how short term greedy, or long term farsighted, the agent should be.

To work out the expected future value for a given agent and environment interacting, we take the sum of these discounted rewards into the infinite future and work out its expected value,

$$V_{\mu}^{\pi}(\gamma) := \frac{1}{\Gamma} \mathbf{E} \left(\sum_{i=1}^{\infty} \gamma^i r_i \right). \quad (1)$$

In the above, r_i is the reward in cycle i of a given history, γ is the discount rate, γ^i is the discount applied to the i^{th} reward into the future, the normalising constant is $\Gamma := \sum_{i=1}^{\infty} \gamma^i$, and the expected value is taken over all possible interaction sequences between the agent π and the environment μ .

Under geometric discounting an agent with $\gamma = 0.95$ will not plan further than about 20 cycles ahead. Thus we say that the agent has a constant effective horizon of $\frac{1}{1-\gamma}$. Since we are interested in universal intelligence, a limited farsightedness is not acceptable because for every horizon there is a task that needs a larger horizon to be solved. For instance, while a horizon of 5 is sufficient for tic-tac-toe, it is insufficient for chess. Clearly, geometric discounting has not solved the problem of how to weight near term rewards versus long term rewards, it has simply expressed this weighting as a parameter. What we require is a single definition of machine intelligence, not a range of definitions that vary according to a free parameter.

A more promising candidate for universal discounting is the near-harmonic, or quadratic discount, where we replace γ^i in Equation 1 by $1/i^2$ and modifying Γ accordingly. This has some interesting properties, in particular the agent needs to look forward into the future in a way that is proportional to its current age. This is appealing since it seems that humans of age k years usually do not plan their lives for more than, perhaps, the next k years. More importantly, it allows us to avoid the problem of having to choose a global time scale or effective horizon [Hut05].

Although harmonic discounting has a number of attractive properties [Hut06a], an even simpler and more general solution is possible. If we look at the value function in Equation 1, we see that discounting plays two roles. Firstly, it normalises rewards received so that their sum is always finite. Secondly, it weights the reward at different points in the future which in effect defines a temporal preference. A direct way to solve both of these problems, without needing an external parameter, is to simply require that the total reward returned by the environment can never exceed 1. For such a reward summable environment μ , it follows that the expected value of the sum of rewards is also finite and thus discounting is no longer required,

$$V_{\mu}^{\pi} := \mathbf{E} \left(\sum_{i=1}^{\infty} r_i \right) \leq 1. \quad (2)$$

One way of viewing this is that the rewards returned by the environment now have the temporal preference already factored in. The cost is that this is an additional condition that we place on the space of environments. Previously we required that each reward signal was in a subset of $[0, 1] \cap \mathbb{Q}$, now we have the additional constraint that the reward sum is always bounded (see Subsection 5.2 for further discussion about why this constraint is reasonable).

Space of environments. Intelligence is not simply the ability to perform well at a narrowly defined task; it is much broader. An intelligent agent is able to adapt and learn to deal with many different situations, kinds of problems and types of environments. In our informal definition this was described as the agent’s general ability to perform well in a “wide range of environments.” This flexibility is a defining characteristic and one of the most important differences between humans and many current AI systems: While Gary Kasparov would still be a formidable player if we were to change the rules of chess, IBM’s Deep Blue chess super computer would be rendered useless without significant human intervention.

As our goal is to produce a definition of intelligence that is as broad and encompassing as possible, the space of environments used in our definition should be as large as possible. As the environment is a probability measure with a certain structure, an obvious possibility would be to consider the space of all probability measures of this form. Unfortunately, this extremely broad class of environments causes serious problems. As the space of all probability measures is uncountably infinite, some environments cannot be described in a finite way and so are uncomputable. This would make it impossible, by definition, to test an agent in such an environment using a computer. Further, most environments would be infinitely complex and have little structure for the agent to learn from.

The solution then, is to require the environmental probability measures to be computable. Not only is this condition necessary if we are to have an effective measure of intelligence, it is also not as restrictive as it might first appear. There are still an infinite number of environments with no upper bound on their maximal complexity. Also, it is only the measure that describes the environment that is computable, and so the way in which the environment responds does not have to be deterministic. For example, although a typical sequence of 1’s and 0’s generated at random by flipping a coin is not computable, the probability measure that describes this distribution is computable and thus it is included in our space of possible environments. Indeed there is currently no evidence that the physical universe cannot be simulated by a Turing machine in the above sense (for further discussion of this point see Subsection 5.2). This appears to be the largest reasonable space of environments.

3.3 A formal definition of machine intelligence

In order to define an overall measure of performance, we need to find a way to combine an agent’s performance in many different environments into a single overall measure. As there are an infinite number of environments, we cannot simply take a uniform distribution over them. Mathematically, we must weight some environments higher than others. But how?

Consider the agent’s perspective on this situation: There exists a probability measure that describes the true environment, however this measure is not known to the agent. The only information the agent has are some past observations of the environment. From these, the agent can construct a list of probability measures that are consistent with the observations. We call these potential explanations of the true environment, hypotheses. As the number of observations increases, the set of hypotheses shrinks and hopefully the remaining hypotheses become increasingly accurate at modelling the true environment.

The problem is that in any given situation there will likely be a large number of hypotheses that are consistent with the current set of observations. Thus, if the agent

is going to predict which hypotheses are the most likely to be correct, it must resort to something other than just the observational information that it has. This is a frequently occurring problem in inductive inference for which the most common approach is to invoke the principle of Occam's razor:

Given multiple hypotheses that are consistent with the data, the simplest should be preferred.

This is generally considered the rational and intelligent thing to do [Wal05], indeed IQ tests often implicitly test an individual's ability to use Occam's razor, as pointed out in Subsection 2.1.

3.3 Example. Consider the following type of question which commonly appears in intelligence tests. There is a sequence such as 2, 4, 6, 8, and the test subject needs to predict the next number. Of course the pattern is immediately clear: The numbers are increasing by 2 each time, or more mathematically, the k^{th} item is given by $2k$. An intelligent person would easily identify this pattern and predict the next digit to be 10. However, the polynomial $2k^4 - 20k^3 + 70k^2 - 98k + 48$ is also consistent with the data, in which case the next number in the sequence would be 58. Why then, even if we are aware of the larger polynomial, do we consider the first answer to be the most likely one? It is because we apply, perhaps unconsciously, the principle of Occam's razor. The fact that intelligence tests define this as the "correct" answer, shows us that using Occam's razor is considered the intelligent thing to do. Thus, although we do not usually mention Occam's razor when defining intelligence, the ability to effectively use it is an important facet of intelligent behaviour.

In some cases we may even consider the correct use of Occam's razor to be a more important demonstration of intelligence than achieving a successful outcome. Consider, for example, the following game:

3.4 Example. A questioner lays twenty \$10 notes out on a table before you and then points to the first one and asks "Yes or No?". If you answer "Yes" he hands you the money. If you answer "No" he takes it from the table and puts it in his pocket. He then points to the next \$10 note on the table and asks the same question. Although you, as an intelligent agent, might experiment with answering both "Yes" and "No" a few times, by the 13th round you would have decided that the best choice seems to be "Yes" each time. However what you do not know is that if you answer "Yes" in the 13th round then the questioner will pull out a gun and shoot you! Thus, although answering "Yes" in the 13th round is the most intelligent choice, given what you know, it is not the most successful one. An exceptionally dim individual may have failed to notice the obvious relationship between answers and getting the money, and thus might answer "No" in the 13th round, thereby saving his life due to what could truly be called "dumb luck".

What is important then, is not that an intelligent agent succeeds in any given situation, but rather that it takes actions that we would expect to be the most likely ones to lead to success. Given adequate experience this might be clear, however often experience is not sufficient and one must fall back on good prior assumptions about the world, such as Occam's razor. It is important then that we test the agents in such a way that they are,

at least on average, rewarded for correctly applying Occam’s razor, even if in some cases this leads to failure.

There is another subtlety that needs to be pointed out. Often intelligence is thought of as the ability to deal with complexity. Or in the words of the psychologist Gottfredson, “... g is the ability to deal with cognitive complexity — in particular, with complex information processing.” [Got97b] It is tempting then to equate the difficulty of an environment with its complexity. Unfortunately, things are not so straightforward. Consider the following environment:

3.5 Example. Imagine a very complex environment with a rich set of relationships between the agent’s actions and observations. The measure that describes this will have a high complexity. However, also imagine that the reward signal is always maximal no matter what the agent does. Thus, although this is a very complex environment in which the agent is unlikely to be able predict what it will observe next, it is also an easy environment in the sense that all policies are optimal, even very simple ones that do nothing at all. The environment contains a lot of structure that is irrelevant to the goal that the agent is trying to achieve.

From this perspective, a problem is thought of as being difficult if the simplest good solution to the problem is complex. Easy problems on the other hand are those that have simple solutions. This is a very natural way to think about the difficulty of problems, or in our terminology, environments.

Fortunately, this distinction does not affect our use of Occam’s razor. When we talk about an hypothesis, what we mean is a potential model of the environment from the agent’s perspective, not just a model that is sufficient with respect to the agent’s goal. From the agent’s perspective, an incorrect hypothesis that fails to model much of the environment may be optimal if the parts of the environment that the hypothesis fails to model are not relevant to receiving reward. However, when Occam’s razor is applied, we apply it with respect to the complexity of the hypotheses, not the complexity of good solutions with respect to an objective. Thus, to reward agents on average for correctly using Occam’s razor, we must weight the environments according to their complexity, not their difficulty.

Our remaining problem now is to measure the complexity of environments. The Kolmogorov complexity of a binary string x is defined as being the length of the shortest program that computes x :

$$K(x) := \min_p \{l(p) : \mathcal{U}(p) = x\},$$

where p is a binary string which we call a *program*, $l(p)$ is the length of this string in bits, and $\mathcal{U}$ is a prefix universal Turing machine $\mathcal{U}$ called the *reference machine*.

To gain an intuition for how this works, consider a binary string $0000 \dots 0$ that consists of a trillion 0s. Although this string is very long, it clearly has a simple structure and thus we would expect it to have a low complexity. Indeed this is the case because we can write a very short program p that simply loops a trillion times outputting a 0 each time. Similarly, other strings with simple patterns have a low Kolmogorov complexity. On the other hand, if we consider a long irregular random string $111010110000010 \dots$ then it is much more difficult to find a short program that outputs this string. Indeed it is possible to prove that there are so many strings of this form, relative to the number

of short programs, that in general it is impossible for long random strings to have short programs. In other words, they have high Kolmogorov complexity.

An important property of K is that it is nearly independent of the choice of $\mathcal{U}$. To see why, consider what happens if we switch from $\mathcal{U}$, in the above definition of K , to some other universal Turing machine $\mathcal{U}'$. Due to the universality property of $\mathcal{U}'$, there exists a program q that allows $\mathcal{U}'$ to simulate $\mathcal{U}$. Thus, if we give $\mathcal{U}'$ both q and p as inputs, it can simulate $\mathcal{U}$ running p and thereby compute $\mathcal{U}(p)$. It follows then that switching from $\mathcal{U}$ to $\mathcal{U}'$ in our definition of K above incurs at *most* an additional cost of $l(q)$ bits in minimal program length. The constant $l(q)$ is independent of which string x we are measuring the complexity of, and for reasonable universal Turing machines, this constant will be small. This invariance property makes K an excellent universal complexity measure. For an extensive treatment of Kolmogorov complexity see [LV97] or [Cal02].

In our current application we need to measure the complexity of the computable measures that describe environments. It can be shown that this set can be enumerated $\mu_1, \mu_2, \mu_3, \dots$ (see Theorem 4.3.1 in [LV97]). Using a simple encoding method we can express each index as a binary string, written $\langle i \rangle$. In a sense this binary string is a description of an environment with respect to our enumeration. This lets us define the complexity of an environment μ_i to be $K(\mu_i) := K(\langle i \rangle)$. Intuitively, if a short program can be used to describe the program for an environment μ_i , then this environment will have a low complexity.

This answers our problem of needing to be able to measure the complexity of environments, but we are not done yet. In order to formalise Occam's razor we need to have a way to assign an a priori probability to environments in such a way that complex environments are less likely, and simple environments more likely. If we consider that each environment μ_i is described by a minimal length program that is a binary string, then the natural way to do this is to consider each additional bit of program length to reduce the environment's probability by one half, reflecting the fact that each bit has two possible states. This gives us what is known as the *algorithmic probability distribution* over the space of environments, defined $2^{-K(\mu)}$. This distribution has powerful properties that essentially solve long-standing open philosophical, statistical, and computational problems in the area of inductive inference [Hut07a]. Furthermore, the distribution can be used to define powerful universal learning agents that have provably optimal performance [Hut05].

Bringing all these pieces together, we can now define our formal measure of intelligence for arbitrary systems. Let E be the space of all computable reward summable environmental measures with respect to the reference machine $\mathcal{U}$, and let K be the Kolmogorov complexity function. The expected performance of agent π with respect to the universal distribution $2^{-K(\mu)}$ over the space of all environments E is given by,

$$\Upsilon(\pi) := \sum_{\mu \in E} 2^{-K(\mu)} V_{\mu}^{\pi}.$$

We call this the *universal intelligence* of agent π .

Consider how this equation corresponds to our informal definition. We needed a measure of an agent's general ability to achieve goals in a wide range of environments. Clearly present in the equation is the agent π , the environment μ and, implicit in the environment, a goal. The agent's "ability to achieve" is represented by the value function V_{μ}^{π} . By a "wide range of environments" we have taken the space of all well defined reward summable environments, where these environments have been characterised as computable measures

in the set E . Occam's razor is given by the term $2^{-K(\mu)}$ which weights the agent's performance in each environment inversely proportional to its complexity. The definition is very general in terms of which sensors or actuators the agent might have as all information exchanged between the agent and the environment takes place over very general communication channels. Finally, the formal definition places no limits on the internal workings of the agent. Thus, we can apply the definition to any system that is able to receive and generate information with view to achieving goals. The main drawback, however, is that the Kolmogorov complexity function K is not computable and can only be approximated. This is an important point that we will return to later.

3.4 Universal intelligence of various agents

In order to gain some intuition for our definition of universal intelligence, in this subsection we will consider a range of different agents and their relative degrees of universal intelligence.

A random agent. The agent with the lowest intelligence, at least among those that are not actively trying to perform badly, would be one that makes uniformly random actions. We will call this π^{rand} . Although this is clearly a weak agent, we cannot simply conclude that the value of $V_{\mu}^{\pi^{\text{rand}}}$ will always be low as some environments will generate high reward no matter what the agent does. Nevertheless, in general such an agent will not be very successful as it will fail to exploit any regularities in the environment, no matter how simple they are. It follows then that the values of $V_{\mu}^{\pi^{\text{rand}}}$ will typically be low compared to other agents, and thus $\Upsilon(\pi^{\text{rand}})$ will be low. Conversely, if $\Upsilon(\pi^{\text{rand}})$ is very low, then the equation for Υ implies that for simple environments, and many complex environments, the value of $V_{\mu}^{\pi^{\text{rand}}}$ must also be relatively low. This kind of poor performance in general is what we would expect of an unintelligent agent.

A very specialised agent. From the equation for Υ , we see that an agent could have very low universal intelligence but still perform extremely well at a few very specific and complex tasks. Consider, for example, IBM's Deep Blue chess supercomputer, which we will represent by π^{dblue} . When μ^{chess} describes the game of chess, $V_{\mu^{\text{chess}}}^{\pi^{\text{dblue}}}$ is very high. However $2^{-K(\mu^{\text{chess}})}$ is small, and for $\mu \neq \mu^{\text{chess}}$ the value function will be low as π^{dblue} only plays chess. Therefore, the value of $\Upsilon(\pi^{\text{dblue}})$ will be very low. Intuitively, this is because Deep Blue is too inflexible and narrow to have general intelligence; a characteristic weakness of specialised artificial intelligence systems.

A general but simple agent. Imagine an agent that performs very basic learning by building up a table of observation and action pairs and keeping statistics on the rewards that follow. Each time an observation that it has been seen before occurs, the agent takes the action with highest estimated expected reward in the next cycle with 90% probability, or a random action with 10% probability. We will call this agent π^{basic} . It is immediately clear that many environments, both complex and very simple, will have at least some structure that such an agent would take advantage of. Thus, for almost all μ we will have $V_{\mu}^{\pi^{\text{basic}}} > V_{\mu}^{\pi^{\text{rand}}}$ and so $\Upsilon(\pi^{\text{basic}}) > \Upsilon(\pi^{\text{rand}})$. Intuitively, this is what we would expect as π^{basic} , while very simplistic, is surely more intelligent than π^{rand} .

Similarly, as π^{dblue} will fail to take advantage of even trivial regularities in some of the most basic environments, $\Upsilon(\pi^{\text{basic}}) > \Upsilon(\pi^{\text{dblue}})$. This is reasonable as our aim is to measure a machine's level of general intelligence. Thus an agent that can take advantage of basic regularities in a wide range of environments should rate more highly than a specialised machine that fails outside of a very limited domain.

A simple agent with more history. The first order structure of π^{basic} , while very general, will miss many simple exploitable regularities. Consider the following environment μ^{alt} . Let $\mathcal{R} = [0, 1] \cap \mathbb{Q}$, $\mathcal{A} = \{\text{up}, \text{down}\}$ and $\mathcal{O} = \{\varepsilon\}$, where ε is the empty string. In cycle k the environment generates a reward of 2^{-k} each time the agent's action is different to its previous action. Otherwise the reward is 0. We can define this environment formally,

$$\mu^{\text{alt}}(o_k r_k | o_1 \dots a_{k-1}) := \begin{cases} 1 & \text{if } a_{k-1} \neq a_{k-2} \wedge r_k = 2^{-k}, \\ 1 & \text{if } a_{k-1} = a_{k-2} \wedge r_k = 0, \\ 0 & \text{otherwise.} \end{cases}$$

Clearly the optimal strategy for an agent is simply to alternate between the actions **up** and **down**. Even though this is very simple, this strategy requires the agent to correlate its current action with its previous action, something that π^{basic} cannot do.

A natural extension of π^{basic} is to use a longer history of actions, observations and rewards in its internal table. Let $\pi^{2\text{back}}$ be the agent that builds a table of statistics for the expected reward conditioned on the last two actions, rewards and observations. It is immediately clear that $\pi^{2\text{back}}$ will exploit the structure of the μ^{alt} environment. Furthermore, by definition $\pi^{2\text{back}}$ is a generalisation of π^{basic} and thus it will adapt to any regularity that π^{basic} can adapt to. It follows then that in general $V_\mu^{\pi^{2\text{back}}} > V_\mu^{\pi^{\text{basic}}}$ and so $\Upsilon(\pi^{2\text{back}}) > \Upsilon(\pi^{\text{basic}})$, as we would intuitively expect. In the same way we can extend the history that the agent utilises back further and produce even more powerful agents that are able to adapt to more lengthy temporal structures and which will have still higher machine intelligence.

A simple forward looking agent. In some environments simply trying to maximise the next reward is not sufficient, the agent must also take into account the rewards that are likely to follow further into the future, that is, the agent must plan ahead. Consider the following environment μ^{slide} . Let $\mathcal{R} = [0, 1] \cap \mathbb{Q}$, $\mathcal{A} = \{\text{rest}, \text{climb}\}$ and $\mathcal{O} = \{\varepsilon\}$. Intuitively, there is a slide such as you would see in a playground. The agent can rest at the bottom of the slide, for which it receives a reward of 2^{-k-4} . The alternative is to climb the slide, which gives a reward of 0. Once at the top of the slide the agent always slides back down no matter what action is taken; this gives a reward of 2^{-k} . This is illustrated in Figure 2. The environment is completely deterministic.

Because climbing receives a reward of 0, while resting receives a reward of 2^{-k-4} , a very shortsighted agent that only tries to maximise the reward in the next cycle will choose to stay at the bottom of the slide. Both π^{basic} and $\pi^{2\text{back}}$ have this problem; though they also take random actions 10% probability and so will occasionally climb the slide by chance. Clearly this is not optimal in terms of total reward over time.

We can extend the $\pi^{2\text{back}}$ agent again by defining a new agent $\pi^{2\text{forward}}$ that with 90% probability chooses its next action to maximise not just the next reward, but $\hat{r}_{k+1} + \hat{r}_{k+2}$, where $\hat{r}_{k+1}$ and $\hat{r}_{k+2}$ are the agent's estimates of the next two rewards. As the estimate

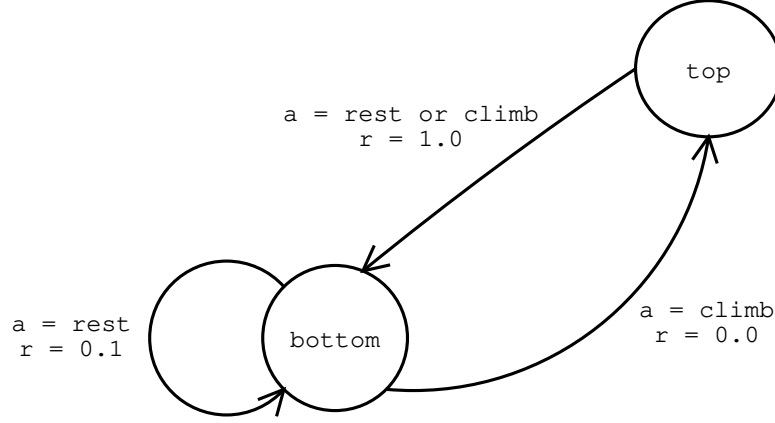


Figure 2: A simple game in which the agent climbs a playground slide and slides back down again. A shortsighted agent will always just rest at the bottom of the slide.

of $\hat{r}_{k+2}$ will potentially depend not only on a_k , but also on a_{k+1} , the agent assumes that a_{k+1} is chosen to simply maximise the estimated reward $\hat{r}_{k+2}$.

The $\pi^{2\text{back}}$ agent can see that by missing out on the resting reward of 2^{-k-4} for one cycle and climbing, a greater reward of 2^{-k} will be had when sliding back down the slide in the following cycle.

By definition $\pi^{2\text{forward}}$ generalises $\pi^{2\text{back}}$ in a way that more closely reflects the value function V and thus in general $V_{\mu}^{\pi^{2\text{forward}}} > V_{\mu}^{\pi^{2\text{back}}}$. It then follows that $\Upsilon(\pi^{2\text{forward}}) > \Upsilon(\pi^{2\text{back}})$ as we would intuitively expect for this more powerful agent.

In a similar way agents of increasing complexity and adaptability can be defined which will have still greater intelligence. However with more complex agents it is usually difficult to theoretically establish whether one agent has more or less universal intelligence than another. Nevertheless, in the simple examples above we saw that the more flexible and powerful an agent was, the higher its universal intelligence.

A very intelligent agent. A very smart agent would perform well in simple environments, and reasonably well compared to most other agents in more complex environments. From the equation for universal intelligence this would clearly produce a very high value for Υ . Conversely, if Υ was very high then the equation for Υ implies that the agent must perform well in most simple environments and reasonably well in many complex ones also.

A super intelligent agent. Consider what would be required to maximise the value of Υ . By definition, a “perfect” agent would always pick the action which had greatest expected future reward. To do this, for every environment $\mu \in E$ the agent must take into account how likely it is that it is facing μ given the interaction history so far, and the prior probability of μ , that is, $2^{-K(\mu)}$. It would then consider all possible future interactions that might occur, and how likely they are, and from this select the action in the current cycle that maximises the expected future reward.

This perfect theoretical agent is known as AIXI. It has been precisely defined and studied at length in [Hut05] (see [Hut07b] for a shorter exposition). The connection between universal intelligence and AIXI is not coincidental: Υ was originally derived from the so called “intelligence order relation” (see Definition 5.14 in [Hut05]), which in

turn was constructed to reflect the equations for AIXI. As such we can define the upper bound on universal intelligence to be,

$$\tilde{\Upsilon} := \max_{\pi} \Upsilon(\pi) = \Upsilon(\pi^{AIXI}).$$

AIXI is not computable due to the incomputability of K , and even if K were computable, accurately computing the expectations to maximise future expected rewards would be practically infeasible. Nevertheless, AIXI is interesting from a theoretical perspective as it defines, in an elegant way, what might be considered to be the perfect theoretical artificial intelligence. Indeed many strong optimality properties have been proven for AIXI. For example, it has been proven that AIXI converges to optimal performance in any environment where this is at all possible for a general agent (see Theorem 5.34 of [Hut05]). This optimality result includes ergodic Markov decision processes, prediction problems, classification problems, bandit problems and many others [LH04b, LH04a]. These mathematical results prove that agents with very high universal intelligence are extremely powerful and general.

A human. For extremely simple environments, a human should be able to identify their simple structure and exploit this to maximise reward. However, for more complex environments it is hard to know how well a human would perform. Much of the human brain is set up to process information from the human sense organs, and thus is quite specialised. Perhaps the amount of universal machine intelligence that a human has is not that high compared to some machine learning algorithms? It is difficult to know without experimental results.

3.5 Properties of universal intelligence

What we have presented is a definition of machine intelligence, it is not a practical test of machine intelligence, indeed the value of Υ is not computable due to the use of Kolmogorov complexity. The difference between the definition of something and practical tests is important to keep in mind. In some cases tests are based on a definition or theory of intelligence. In other cases, as we will see in the next section, what is presented is some where between a fully encompassing definition, and a realistically practical test. Thus the distinction between tests and definitions is not always clear.

Here our goal has simply been to define the concept of machine intelligence in the most general, powerful and elegant way. In future research we will explore ways to approximate this ideal with a practical test. Naturally the process of estimation will introduce weaknesses and flaws that the original definition did not have. For example, while the definition considers the general performance of an agent over all computable environments with bounded reward sum, in practice a test could only ever estimate this by testing the agent on a finite sample of environments.

A similar situation arises when defining randomness for sequences. In essence, we consider an infinite sequence to be Martin-Löf random when it has no significant regularity. This lack of regularity is equivalent to saying that the sequence cannot be compressed in any significant way, and thus we can characterise randomness using Kolmogorov complexity. Naturally, we cannot test a sequence for every possible regularity, which is equivalent to saying that we cannot compute its Kolmogorov complexity. We can however test sequences for randomness by checking them for a large number of statistical regularities,

indeed this is what is done in practice. Of course, just because a sequence passes all our tests does not mean that it must be random. There could always be some deeper structure to the sequence that our tests were not able to detect. All we can say is that the sequence seems random with respect to our ability to detect patterns.

Some might argue that the definition of something should not just capture the concept, it should also be practical. For example, the definition of intelligence should be such that intelligence can be easily measured. The above example, however, illustrates why this approach is sometimes flawed: If we were to *define* randomness with respect to a particular set of tests, then one could specifically construct a sequence that followed a regular pattern in such a way that it passed all of our randomness tests. This would completely undermine our definition of randomness. A better approach is to define the concept in the strongest and cleanest way possible, and then to accept that our ability to test for this ideal has limitations. In other words, our task is to find better and more effective tests, not to redefine what it is that we are testing for. This is the attitude we have taken here, though in this paper our focus is almost entirely on the first part, that is, establishing a strong theoretical definition of machine intelligence.

Although some of the criteria by which we judge practical tests of intelligence are not relevant to a pure definition of intelligence, many of the desirable properties are similar. Thus to understand the strengths and weaknesses of our definition, consider again the desirable properties for a test of intelligence from Subsection 2.3.

Valid. The most important property of any proposed formal definition of intelligence is that it does indeed describe something that can reasonably be called “intelligence”. Essentially, this is the core argument of this report so far: We have taken a mainstream informal definition and step by step formalised it. Thus, so long as our informal definition is reasonable, and our formalisation argument holds, the result can reasonably be described as a formal definition of intelligence.

Meaningful. As we saw in the previous section, universal intelligence orders the power and adaptability of simple agents in a natural way. Furthermore, a high value of Υ implies that the agent performs well in most simple and moderately complex environments. Such an agent would be an impressively powerful and flexible piece of technology, with many potential uses. Clearly then, universal intelligence is inherently meaningful, independent of whether or not one considers it to be a measure of intelligence.

Informative. $\Upsilon(\pi)$ is a real value that is independent of the performance of other possible agents. Thus we can make direct comparisons between different agents on a single scale. This property is important if we want to use this measure to study new algorithms.

Wide range. As we saw in the previous section, universal intelligence is able to order the intelligence of even the most basic agents such as π^{rand} , π^{basic} , π^{2back} and π^{2forward} . At the other extreme we have the theoretical super intelligent agent AIXI which has maximal Υ value. Thus, universal intelligence spans trivial learning algorithms right up to super intelligent agents. This seems to be the widest range possible for a measure of machine intelligence.

General. As the agent’s performance on all well defined environments is factored into its Υ value, a broader performance metric is difficult to imagine. Indeed, a well defined measure of intelligence that is broader than universal intelligence would seem to contradict the Church-Turing thesis as it would imply that we could effectively measure an agent’s performance for some well defined problem that was outside of the space of computable measures.

Unbiased. In a standard intelligence test, an individual’s performance is judged on specific kinds of problems, and then these scores are combined to produce an overall result. Thus a test’s outcome is a product of which types of problems it uses and how each score is weighted to produce the end result. Unfortunately, how we do this is a product of many things, including our culture, values and the theoretical perspective on intelligence that we have taken. For example, while one intelligence test could contain many logical puzzle problems, another might be more linguistic in emphasis, while another stresses visual reasoning. Modern intelligence tests like the Stanford-Binet try to minimise this problem by covering the most important areas of human reasoning both verbally and non-verbally. This helps but it is still very anthropocentric as we are still only testing those abilities that we think are important for human intelligence.

For an intelligence measure for arbitrary machines we have to base the test on something more general and principled: Universal Turing computation. As all proposed models of computation have thus far been equivalent in their expressive power, the concept of computation appears to be a fundamental theoretical property rather than the product of any specific culture. Thus, by weighting different environments depending on their Kolmogorov complexity, and considering the space of all computable environments, we have avoided having to define intelligence with respect to any particular culture, species etc.

Unfortunately, we have not entirely removed the problem. The environmental distribution $2^{-K(\mu)}$ that we have used is invariant, up to a multiplicative constant, to changes in the reference machine $\mathcal{U}$. Although this affords us some protection, the relative intelligence of agents can change if we change our reference machine. One approach to this problem is to limit the complexity of the reference machine, for example by limiting its state-symbol complexity. We expect that for highly intelligent machines that can deal with a wide range of environments of varying complexity, the effect of changing from one simple reference machine to another will be minor. For simple agents, such as those considered in Subsection 3.4, the ordering of their machine intelligence was also not particularly sensitive to natural choices of reference machine. Recently attempts have been made to make algorithmic probability completely unique and objective by identifying which universal Turing machines are, in some sense, the most simple [Mül06]. Unfortunately however, an elegant solution to this problem has not yet been found.

Fundamental. Universal intelligence is based on computation, information and complexity. These are fundamental concepts that seem unlikely to change in the future with changes in technology. Indeed, if they were to change, the implications would drastically affect the entire field of computer science, not just this work.

Formal. Universal intelligence is expressed as a mathematical equation and thus there is little space for ambiguity in the definition.

Objective. Universal intelligence does not depend on any subjective criteria.

Universal. Universal intelligence is in no way anthropocentric.

Practical. In its current form the definition cannot be directly turned into a test of intelligence as the Kolmogorov complexity function is not computable. Thus in its pure form we can only use it to analyse the nature of intelligence and to theoretically examine the intelligence of mathematically defined learning algorithms.

In order to use universal intelligence more generally we will need to construct a workable test that approximates an agent's Υ value. The equation for Υ suggests how we might approach this problem. Essentially, an agent's universal intelligence is a weighted sum of its performance over the space of all environments. Thus, we could randomly generate programs that describe environmental probability measures and then test the agent's performance against each of these environments. After sampling sufficiently many environments the agent's approximate universal intelligence would be computed by weighting its score in each environment according to the complexity of the environment as given by the length of its program. Another possibility might be to try to approximate the sum by enumerating environmental programs from short to long, as the short ones will contribute by far the greatest to the sum. However in this case we will need to be able to reset the state of the agent so that it cannot cheat by learning our environmental enumeration method. In any case, various practical challenges will need to be addressed before universal intelligence can be used to construct an effective intelligence test. As this would be a significant project in its own right, in this paper we focus on the theoretical issues surrounding the universal intelligence.

4 Definitions and Tests of Machine Intelligence

In this section we will survey both definitions and tests of machine intelligence. We begin with a sample of informal definitions of machine intelligence. A comprehensive survey of informal definitions is practically impossible as they often appear buried deep in articles or books. Nevertheless we have attempted to collect as many as possible and present a sample of some of the more common perspectives that have been taken.

We then survey formal definitions and tests of machine intelligence. As we will see, it is not always clear whether a proposal is a test, a formal definition, or something in between. In some cases the authors claim it is one or the other, and in some cases both. More accurately there is a spectrum of possibilities and thus we will not attempt to artificially divide them into either tests or formal definitions.

To the best of our knowledge, this section is the only general survey of tests and definitions of machine intelligence. This is remarkable given that the definition and measurement of intelligence in machines are two of the most fundamental questions in artificial intelligence. Currently most text books say very little about intelligence, other than mentioning the Turing test. We hope that our short survey will help to raise awareness of the many other proposals.

4.1 Informal definitions of machine intelligence

The following sample of informal definitions of machine intelligence capture a range of perspectives. For a more comprehensive list of definitions, visit our full collection online [LH07]. We begin with five definitions that have clear connections to our informal definition:

“...the mental ability to sustain successful life.” K. Warwick quoted in [Aso03]

“...doing well at a broad range of tasks is an empirical definition of ‘intelligence’ ” H. Masum [MCO02]

“Intelligence is the computational part of the ability to achieve goals in the world. Varying kinds and degrees of intelligence occur in people, many animals and some machines.” J. McCarthy [McC04]

“Any system ...that generates adaptive behaviour to meet goals in a range of environments can be said to be intelligent.” D. Fogel [Fog95]

“...the ability of a system to act appropriately in an uncertain environment, where appropriate action is that which increases the probability of success, and success is the achievement of behavioral subgoals that support the system’s ultimate goal.” J. S. Albus [Alb91]

The position taken by Albus is especially similar to ours. Although the quote above does not explicitly mention the need to be able to perform well in a wide range of environments, at a later point in the same paper he mentions the need to be able to succeed in a “large variety of circumstances”.

“Intelligent systems are expected to work, and work well, in many different environments. Their property of intelligence allows them to maximize the probability of success even if full knowledge of the situation is not available. Functioning of intelligent systems cannot be considered separately from the environment and the concrete situation including the goal.” R. R. Gudwin [Gud00]

While this definition is consistent with the position we have taken, when trying to actually test the intelligence of an agent Gudwin does not believe that a “black box” behaviour based approach is sufficient, rather his approach is to look at the “...architectural details of structures, organizations, processes and algorithms used in the construction of the intelligent systems,” [Gud00] Our perspective is simply to not care whether an agent looks intelligent on the inside. If it is able to perform well in a wide range of environments, that is all that matters. For more discussion on this point see our response to Block’s and Searle’s arguments in Subsection 5.2.

“We define two perspectives on artificial system intelligence: (1) native intelligence, expressed in the specified complexity inherent in the information content of the system, and (2) performance intelligence, expressed in the successful (i.e., goal-achieving) performance of the system in a complicated environment.” J. A. Horst [Hor02]

Here we see two distinct notions of intelligence, a performance based one and an information content one. This is similar to the distinction between fluid intelligence and crystallized intelligence made by the psychologist Cattell (see Subsection 2.5). The performance notion of intelligence is similar to our definition with the expectation that performance is measured in a complex environment rather than across a wide range of environments. This perspective appears in some other definitions also,

“... the ability to solve hard problems.” M. Minsky [Min85]

“Achieving complex goals in complex environments” B. Goertzel [Goe06]

Interestingly, Goertzel claims that an AI system he is developing should be able to, with sufficient resources, perform arbitrarily well with respect to the intelligence order relation, that is, the relation on which universal intelligence was originally based [LGP04]. Presumably then he does not consider his definition to be significantly incompatible with ours.

Some definitions emphasise not just the ability to perform well, but also the need for efficiency:

“[An intelligent agent does what] is appropriate for its circumstances and its goal, it is flexible to changing environments and changing goals, it learns from experience, and it makes appropriate choices given perceptual limitations and finite computation.” D. Poole [PMG98]

“... in any real situation behavior appropriate to the ends of the system and adaptive to the demands of the environment can occur, within some limits of speed and complexity.” A. Newell and H. A. Simon [NS76]

“Intelligence is the ability to use optimally limited resources – including time – to achieve goals.” R. Kurzweil [Kur00]

“Intelligence is the ability for an information processing agent to adapt to its environment with insufficient knowledge and resources.” P. Wang [Wan95]

We consider the addition of resource limitations to the definition of intelligence to be either superfluous, or wrong. In the first case, if limited computational resources are a fundamental and unavoidable part of reality, which certainly seems to be the case, then their addition to the definition of intelligence is unnecessary. Perhaps the first three definitions above fall into this category.

On the other hand, if limited resources are not a fundamental restriction, for example a new model of computation was discovered that was vastly more powerful than the current model, then it would be odd to claim that the unbelievably powerful machines that would then result were not intelligence. Normally we do not judge the intelligence of something relative to the resources it uses. For example, if a rat had human level learning and problem solving abilities, we would not think of the rat as being more intelligence than a human due to the fact that its brain was much smaller.

While we do not consider efficiency to be a part of the definition of intelligence, this is not to say that considering the efficiency of agents is unimportant. Indeed, a key goal of

artificial intelligence is to find algorithms which have the greatest efficiency of intelligence, that is, which achieve the most intelligence per unit of computational resources consumed.

It should also be pointed out that although universal intelligence does not test the efficiency of an agent in terms of the computational resources that it uses, it does however test how quickly the agent learns from past data. In a sense, an agent which learns very quickly could be thought of as being very “data efficient”.

4.2 Formal definitions and tests of machine intelligence

Turing test and derivatives. The classic approach to determining whether a machine is intelligent is the so called Turing test [Tur50] which has been extensively debated over the last 50 years [SCA00]. Turing realised how difficult it would be to directly define intelligence and thus attempted to side step the issue by setting up his now famous imitation game: If human judges can not effectively discriminate between a computer and a human through teletyped conversation then we must conclude that the computer is intelligent.

Though simple and clever, the test has attracted much criticism. Block and Searle argue that passing the test is not *sufficient* to establish intelligence [Blo81, Sea80, Eis91]. Essentially they both argue that a machine could appear to be intelligent without having any “real intelligence”, perhaps by using a very large table of answers to questions. While such a machine might be impossible in practice due to the vast size of the table required, it is not logically impossible. In which case an unintelligent machine could, at least in theory, consistently pass the Turing test. Some consider this to bring the validity of the test into question. In response to these challenges, even more demanding versions of the Turing test have been proposed such as the Total Turing test [Har89], the Truly Total Turing test [Sch98] and the inverted Turing test [Wat96]. Dowe argues that the Turing test should be extended by ensuring that the agent has a compressed representation of the domain area, thus ruling out look-up table counter arguments [DH98]. Of course these attacks on the Turing test can be applied to any test of intelligence that considers only a system’s external behaviour, that is, most intelligence tests.

A more common criticism is that passing the Turing test is not *necessary* to establish intelligence. Usually this argument is based on the fact that the test requires the machine to have a highly detailed model of human knowledge and patterns of thought, making it a test of humanness rather than intelligence [Fre90, FH98]. Indeed, even small things like pretending to be *unable* to perform complex arithmetic quickly and faking human typing errors become important, something which clearly goes against the purpose of the test.

The Turing test has other problems as well. Current AI systems are a long way from being able to pass an unrestricted Turing test. From a practical point of view this means that the full Turing test is unable to offer much guidance to our work. Indeed, even though the Turing test is the most famous test of machine intelligence, almost no current research in artificial intelligence is specifically directed toward being able to pass it. Unfortunately, simply restricting the domain of conversation in the Turing test to make the test easier, as is done in the Loebner competition [Loe90], is not sufficient. With restricted conversation possibilities the most successful Loebner entrants are even more focused on faking human fallibility, rather than anything resembling intelligence [Hut96]. Finally, the Turing test returns different results depending on who the human judges are. Its unreliability has in some cases lead to clearly unintelligent machines being classified as human, and at least

one instance of a human actually failing a Turing test. When queried about the latter, one of the judges explained that “no human being would have that amount of knowledge about Shakespeare” [Shi94].

Compression tests. Mahoney has proposed a particularly simple solution to the binary pass or fail problem with the Turing test: Replace the Turing test with a text compression test [Mah99]. In essence this is somewhat similar to a “Cloze test” where an individual’s comprehension and knowledge in a domain is estimated by having them guess missing words from a passage of text.

While simple text compression can be performed with symbol frequencies, the resulting compression is relatively poor. By using more complex models that capture higher level features such as aspects of grammar, the best compressors are able to compress text to about 1.5 bits per character for English. However humans, which can also make use of general world knowledge, the logical structure of the argument etc., are able to reduce this down to about 1 bit per character. Thus the compression statistic provides an easily computed measure of how complete a machine’s models of language, reasoning and domain knowledge are, relative to a human.

To see the connection to the Turing test, consider a compression test based on a very large corpus of dialogue. If a compressor could perform extremely well on such a test, this is mathematically equivalent to being able to determine which sentences are probable at a give point in a dialogue, and which are not (for the equivalence of compression and prediction see [BCW90]). Thus, as failing a Turing test occurs when a machine (or person!) generates a sentence which would be improbable for a human, extremely good performance on dialogue compression implies the ability to pass a Turing test.

A recent development in this area is the Hutter Prize [Hut06b]. In this test the corpus is a 100 MB extract from Wikipedia. The idea is that this should represent a reasonable sample of world knowledge and thus any compressor that can perform very well on this test must have have a good model of not just English, but also world knowledge in general.

One criticism of compression tests is that it is not clear whether a powerful compressor would easily translate into a general purpose artificial intelligence. Also, while a young child has a significant amount of elementary knowledge about how to interact with the world, this knowledge would be of little use when trying to compress an encyclopedia full of abstract “adult knowledge” about the world.

Linguistic complexity. A more linguistic approach is taken by the HAL project at the company Artificial Intelligence NV [TGH01]. They propose to measure a system’s level of conversational ability by using techniques developed to measure the linguistic ability of children. These methods examine things such as vocabulary size, length of utterances, response types, syntactic complexity and so on. This would allow systems to be “...assigned an age or a maturity level beside their binary Turing test assessment of ‘intelligent’ or ‘not intelligent’ ” [TGDH00]. As they consider communication to be the basis of intelligence, and the Turing test to be a valid test of machine intelligence, in their view the best way to develop intelligence is to retrace the way in which human linguistic development occurs. Although they do not explicitly refer to their linguistic measure as a test of intelligence, because it measures progress towards what they consider to be a valid intelligence test, it acts as one.

Multiple cognitive abilities. A broader developmental approach is being taken by IBM’s Joshua Blue project [AABL02]. In this project they measure the performance of their system by considering a broad range of linguistic, social, association and learning tests. Their goal is to first pass what they call a “toddler Turing test”, that is, to develop an AI system that can pass as a young child in a similar set up to the Turing test.

Another company pursuing a similar developmental approach based on measuring system performance through a broad range of cognitive tests is the a2i2 project at Adaptive AI [Vos05]. Rather than toddler level intelligence, their current goal is to work toward a level of cognitive performance similar to that of a small mammal. The idea being that even a small mammal has many of the key cognitive abilities required for human level intelligence working together in an integrated way.

Competitive games. The Turing Ratio method of Masum et al. has more emphasis on tasks and games rather than cognitive tests. Similar to our own definition, they propose that “...doing well at a broad range of tasks is an empirical definition of ‘intelligence’.” [MCO02] To quantify this they seek to identify tasks that measure important abilities, admit a series of strategies that are qualitatively different, and are reproducible and relevant over an extended period of time. They suggest a system of measuring performance through pairwise comparisons between AI systems that is similar to that used to rate players in the international chess rating system. The key difficulty however, which the authors acknowledge is an open challenge, is to work out what these tasks should be, and to quantify just how broad, important and relevant each is. In our view these are some of the most central problems that must be solved when attempting to construct an intelligence test. Thus we consider this approach to be incomplete in its current state.

Collection of psychometric tests. An approach called Psychometric AI tries to address the problem of what to test for in a pragmatic way. In the view of Bringsjord and Schimanski, “Some agent is intelligent if and only if it excels at all established, validated tests of [human] intelligence.” [BS03] They later broaden this to also include “tests of artistic and literary creativity, mechanical ability, and so on.” With this as their goal, their research is focused on building robots that can perform well on standard psychometric tests designed for humans, such as the Wechsler Adult Intelligent Scale and Raven Progressive Matrices (see Subsection 2.1).

As effective as these tests are for humans, we believe that they are unlikely to be adequate for measuring machine intelligence. For a start they are highly anthropocentric. Another problem is that they embody basic assumptions about the test subject that are likely to be violated by computers. For example, consider the fundamental assumption that the test subject is not simply a collection of specialised algorithms designed only for answering common IQ test questions. While this is obviously true of a human, or even an ape, it may not be true of a computer. The computer could be nothing more than a collection of specific algorithms designed to identify patterns in shapes, predict number sequences, write poems on a given subject or solve verbal analogy problems — all things that AI researchers have worked on. Such a machine might be able to obtain a respectable IQ score [SD03], even though outside of these specific test problems it would be next to useless. If we try to correct for these limitations by expanding beyond standard tests, as Bringsjord and Schimanski seem to suggest, this once again opens up the difficulty of

exactly what, and what not, to test for. Thus we consider Psychometric AI, at least as it is currently formulated, to only partially address this central question.

C-Test. One perspective among psychologists who support the *g*-factor view of intelligence, is that intelligence is “the ability to deal with complexity”[Got97b]. Thus, in a test of intelligence, the most difficult questions are the ones that are the most complex because these will, by definition, require the most intelligence to solve. It follows then that if we could formally define and measure the complexity of test problems using complexity theory we could construct a formal test of intelligence. The possibility of doing this was perhaps first suggested by Chaitin [Cha82]. While this path requires numerous difficulties to be dealt with, we believe that it is the most natural and offers many advantages: It is formally motivated, precisely defined and potentially could be used to measure the performance of both computers and biological systems on the same scale without the problem of bias towards any particular species or culture.

Essentially this is the approach that we have taken. Universal intelligence is based on the universally optimal AIXI agent for active environments, which in turn is based on Kolmogorov complexity and Solomonoff’s universal model of sequence prediction. A relative of universal intelligence is the C-Test of Hernández-Orallo which was also inspired by Solomonoff induction and Kolmogorov complexity [HO00b, HOMC98]. If we gloss over some technicalities, the essential relationships look like this:

	<i>Universal agent</i>	<i>Universal test</i>
<i>Passive environment</i>	Solomonoff induction	C-Test
<i>Active environment</i>	AIXI	Universal intelligence

The C-Test consists of a number of sequence prediction and abduction problems similar to those that appear in many standard IQ tests. The test has been successfully applied to humans with intuitively reasonable results [HOMC98, HO00a]. Similar to standard IQ tests, the C-Test always ensures that each question has an unambiguous answer in the sense that there is always one hypothesis that is consistent with the observed pattern that has significantly lower complexity than the alternatives. Other than making the test easier to score, it has the added advantage of reducing the test’s sensitivity to changes in the reference machine.

The key difference to sequence problems that appear in standard intelligence tests is that the questions are based on a formally expressed measure of complexity. To overcome the problem of Kolmogorov complexity not being computable, the C-Test instead uses Levin’s *Kt* complexity [Lev73]. In order to retain the invariance property of Kolmogorov complexity, Levin complexity requires the additional assumption that the universal Turing machines are able to simulate each other in linear time. As far as we know, this is the only formal definition of intelligence that has so far produced a usable test of intelligence.

To illustrate the C-Test, below are some example problems taken from [HOMC98]. Beside each question is its complexity, naturally more complex patterns are also more difficult:

<i>Sequence Prediction Test</i>		
Complexity	Sequence	Answer
9	a, d, g, j, _ , ...	m
12	a, a, z, c, y, e, x, _ , ...	g
14	c, a, b, d, b, c, c, e, c, d, _ , ...	d

<i>Sequence Abduction Test</i>		
Complexity	Sequence	Answer
8	a, _ , a, z, a, y, a, ...	a
10	a, x, _ , v, w, t, u, ...	y
13	a, y, w, _ , w, u, w, u, s, ...	y

Our main criticism of the C-Test is that it is a static test limited to passive environments. As we have argued earlier, we believe that a better approach is to use dynamic intelligence tests where the agent must interact with an environment in order to solve problems. As AIXI is a generalisation of Solomonoff induction from passive to active environments, universal intelligence could be viewed as generalising the C-Test from passive to active environments.

Smith’s Test. Another complexity based formal definition of intelligence that appeared recently in an unpublished report is due to W. D. Smith [Smi06]. His approach has a number of connections to our work, indeed Smith states that his work is largely a “...rediscovery of recent work by Marcus Hutter”. Perhaps this is over stating the similarities because while there are some connections, there are also many important differences.

The basic structure of Smith’s definition is that an agent faces a series of problems that are generated by an algorithm. In each iteration the agent must try to produce the correct response to the problem that it has been given. The problem generator then responds with a score of how good the agent’s answer was. If the agent so desires it can submit another answer to the same problem. At some point the agent requests to the problem generator to move onto the next problem and the score that the agent received for its last answer to the current problem is then added to its cumulative score. Each interaction cycle counts as one time step and the agent’s intelligence is then its total cumulative score considered as a function of time. In order to keep things feasible, the problems must all be in the complexity class P, that is, decision problems which can be solved by a deterministic Turing machine in polynomial time.

We have three main criticisms of Smith’s definition. Firstly, while for practical reasons it might make sense to restrict problems to be in P, we do not see why this practical restriction should be a part of the very definition of intelligence. If some breakthrough meant that agents could solve difficult problems in not just P but sometimes in NP as well, then surely these new agents would be more intelligent? We had similar objections to informal definitions of machine intelligence that included efficiency requirements in Subsection 4.1.

Our second criticism is that the way intelligence is measured is essentially static, that is, the environments are passive. As we have argued before, we believe that dynamic testing in active environments is a better measure of a system’s intelligence. To put this

argument yet another way: Succeeding in the real world requires you to be more than an insightful spectator!

The final criticism is that while the definition is somewhat formally defined, still it leaves open the important question of what exactly the tests should be. Smith suggests that researchers should dream up tests and then contribute them to some common pool of tests. As such, this is not a fully specified definition.

4.3 Comparison of machine intelligence tests and definitions

In order to compare the machine intelligence tests and definitions in the previous section, we return again to the desirable properties of a test of intelligence.

Each property is briefly defined followed by a summary comparison in Table 1. Although we have attempted to be as fair as possible, some of the scores we give on this table will be debatable. Nevertheless, we hope that it provides a rough overview of the relative strengths and weaknesses of the proposals.

Valid. A test/measure of intelligence should be just that, it should capture intelligence and not some related quantity or only a part of intelligence.

Informative. The result should be a scalar value, or perhaps a vector, depending on our view of intelligence. We would like an absolute measure of intelligence so that comparisons across many agents can easily be made.

Wide range. A test/definition should cover very low levels of intelligence right up to super human intelligence.

General. Ideally we would like to have a very general test/definition that could be applied to everything from a fly to a machine learning algorithm.

Dynamic. A test/definition should directly take into account the ability to learn and adapt over time as this is an important aspect of intelligence.

Unbiased. A test/definition should not be biased towards any particular culture, species, etc.

Fundamental. We do not want a test/definition that needs to be changed from time to time due to changing technology and knowledge.

Formal. The test/definition should be specified with the highest degree of precision possible, allowing no room for misinterpretation. Ideally, it should be described using formal mathematics.

Objective. The test/definition should not appeal to subjective assessments such as the opinions of human judges.

Fully Defined. Has the test/definition been fully defined, or are parts still unspecified?

Universal. Is the test/definition universal, or is it anthropocentric?

Practical. A test should be able to be performed quickly and automatically, while from a definition it should be possible to create an efficient test.

Intelligence Test	Valid	Informative	Wide Range	General	Dynamic	Unbiased	Fundamental	Formal	Objective	Fully Defined	Universal	Practical	Test vs. Def.
Turing Test	•	•	•	•	•	•	•	•	•	•	•	•	T
Total Turing Test	•	•	•	•	•	•	•	•	•	•	•	•	T
Inverted Turing Test	•	•	•	•	•	•	•	•	•	•	•	•	T
Toddler Turing Test	•	•	•	•	•	•	•	•	•	•	•	•	T
Linguistic Complexity	•	•	•	•	•	•	•	•	•	•	•	•	T
Text Compression Test	•	•	•	•	•	•	•	•	•	•	•	•	T
Turing Ratio	•	•	•	•	•	•	•	•	•	•	•	•	T/D
Psychometric AI	•	•	•	•	•	•	•	•	•	•	•	•	T/D
Smith’s Test	•	•	•	•	•	•	•	•	•	•	•	•	T/D
C-Test	•	•	•	•	•	•	•	•	•	•	•	•	T/D
Universal Intelligence	•	•	•	•	•	•	•	•	•	•	•	•	D

Table 1: In the table ● means “yes”, • means “debatable”, · means “no”, and ? means unknown. When something is rated as unknown that is usually because the test in question is not sufficiently specified.

Test vs. Def. Finally, we note whether the proposal is more of a test, more of a definition, or something in between.

5 Discussion and Conclusions

5.1 Constructing a test of universal intelligence

The central challenge for future work on universal intelligence is to convert the theoretical definition of machine intelligence presented in this paper into a workable test. The basic structure of such a test is already apparent from the equation for Υ : The test would work by evaluating the performance of an agent on a large sample of simulated environments, and then combining the agent’s performance in each environment into an overall intelligence value. This would be done by weighting the agent’s performance in each environment according to the environment’s complexity.

The key theoretical challenge that will need to be deal with is to find a suitable replacement for the incomputable Kolmogorov complexity function. One solution could be to use Levin’s Kt complexity [Lev73], another might be to use Schmidhuber’s Speed prior [Sch02]. Both of these consider the complexity of an algorithm to be determined by both its minimal description length and running time. This forces the complexity measures to be computable. Taking computation time into account also makes reasonable intuitive sense because we would not usually consider a very short algorithm that takes an enormous amount of time to run to be a particularly simple one. The fact that such an approach can be made to work is evidenced by the C-Test.

5.2 Response to common criticisms

What we have attempted to do is very ambitious and so, not surprisingly, the reactions we get can be interesting. Having presented the essence of this work as posters at several conferences, and also as a 30 minute talk, we now have some idea of what the typical responses are. Most people start out skeptical but end up generally enthusiastic, even if they still have a few reservations. This positive feedback has helped motivate us to continue this direction of research. In this subsection, however, we will attempted to cover some of the more common criticisms.

It’s obviously false, there’s nothing in your definition, just a few equations.

Perhaps the most common criticism is also the most vacuous one: It’s obviously wrong! These people seem to believe that defining intelligence with an equation is clearly impossible, and thus there must be very large and obvious flaws in our work. Not surprisingly these people are also the least likely to want to spend 10 minutes having the material explained to them. Unfortunately, none of these people have been able to communicate why the work is so obviously flawed in any concrete way — despite in one instance having one of the authors chasing the poor fellow out of the conference centre and down the street begging for an explanation. If anyone would like to properly explain their position to us in the future, we promise not to chase you down the street!

It’s obviously correct, indeed everybody already knows this stuff. Curiously, the second most common criticism is the exact opposite: The work is obviously right, and indeed it is already well known. Digging deeper, the heart of this criticism comes from the perception that we have not done much more than just describe reinforcement learning. If you already accept that the reinforcement learning framework is the most general and flexible way to describe artificial intelligence, and not everybody does, then by mixing in Occam’s razor and a dash of complexity theory, the equation for universal intelligence follows in a fairly straightforward way. While this is true, the way in which we have brought these things together has never been done before, although it does have some connection to other work, as discussed in Subsection 4.2. Furthermore, simply coming up with an equation is not enough, one must argue that what the equation describes is in fact “intelligence” in a sense that is reasonable for machines.

We have addressed this question in three main ways: Firstly, in Section 2 we developed an informal definition of intelligence based on expert definitions which was then piece by piece formalised leading to the equation for Υ in Subsection 3.3. This chain of argument strongly ties our equation for intelligence with existing informal definitions and ideas on the nature of intelligence. Secondly, in Subsections 3.4 and 3.5 we showed that the equation has properties that are consistent with a definition of intelligence. Finally, in Subsection 3.4 it was shown that universal intelligence is strongly connected to the theory of universally optimal learning agents, in particular AIXI. From this it follows that machines with very high universal intelligence have a wide range of powerful optimality properties. Clearly then, what we have done goes far beyond merely restating elementary reinforcement learning theory.

Assuming that the environment is computable is too strong. It is certainly possible that the physical universe is not computable, in the sense that the probability

distribution over future events cannot, even in theory, be simulated to an arbitrary precision by a computable process. Some people take this position on various philosophical grounds, such as the need for freewill. However, in standard physics there is no law of the universe that is not computable in the above sense. Nor is there any experimental evidence showing that such a physical law must exist. This includes quantum theory and chaotic systems, both of which can be extremely difficult to compute for some physical systems, but are not fundamentally incomputable theories. In the case of quantum computers, they can compute with lower time complexity than classical Turing machines, however they are unable to compute anything that a classical Turing machine cannot, when given enough time. Thus, as there is no hard evidence of incomputable processes in the universe, our assumption that the agent's environment has a computable distribution is certainly not unreasonable.

If a physical process was ever discovered that was not Turing computable, then this would likely result in a new extended model of computation. Just as we have based universal intelligence on the Turing model of computation, it might be possible to construct a new definition of universal intelligence based on this new model in a natural way.

Finally, even if the universe was not computable, and we did not update our formal definition of intelligence to take this into account, the fact that everything in physics so far is computable means that a computable approximation to our universe would still be extremely accurate over a huge range of situations. In which case, an agent that could deal with a wide range of computable environments would most likely still function well within such a universe.

Assuming that environments return bounded sum rewards is unrealistic. If an environment μ is an artificial game, like chess, then it seems fairly natural for μ to meet any requirements in its definition, such as having a bounded reward sum. However if we think of the environment μ as being the universe in which the agent lives, then it seems unreasonable to expect that it should be required to respect such a bound.

Strictly speaking, reward is an interpretation of the state of the environment. In this case the environment is the universe, and clearly the universe does not have any notion of reward for particular agents. In humans this interpretation is internal, for example, the pain that is experienced when you touch something hot. In which case, maybe it should really be a part of the agent rather than the environment? If we gave the agent complete control over rewards then our framework would become meaningless: the perfect agent could simply give itself constant maximum reward. Perhaps the analogous situation for humans would be taking drugs.

A more accurate framework would consist of an agent, an environment and a separate goal system that interpreted the state of the environment and rewarded the agent appropriately. In such a set up the bounded rewards restriction would be a part of the goal system and thus the above philosophical problem would not occur. However, for our current purposes, it is sufficient just to fold this goal mechanism into the environment and add an easily implemented constraint to how the environment may generate rewards. One simple way to bound an environment's total rewards would be to use geometric or harmonic discounting.

How do you respond to Block's "Blockhead" argument? The approach we have taken is unabashedly functional. Theoretically, we desired to have a formal, simple and

very general definition. This is easier to do if we abstract over the internal workings of the agent and define intelligence only in terms of external communications. Practically, what matters is *how well something works*. By definition, if an agent has a high value of Υ , then it must work well over a wide range of environments.

Block attacks this perspective by describing a machine that appears to be intelligent as it is able to pass the Turing test, but is in fact no more than just a big look-up table of questions and answers [Blo81] (for a related argument see [Gun71]). Although such a look-up table based machine would be unfeasibly large, the fact that a finite machine could in theory consistently pass the Turing test, seemingly without any real intelligence, is worrisome. Our formal measure of machine intelligence could be challenged in the same way, as could any test of intelligence that relies only on an agent's external behaviour.

Our response to this is very simple: If an agent has a very high value of Υ then it is, by definition, able to successfully operate in a wide range of environments. We simply do not care whether the agent is efficient, due to some very clever algorithm, or absurdly inefficient, for example by using an unfeasibly gigantic look-up table of precomputed answers. The important point for us is that the machine has an amazing ability to solve a huge range of problems in a wide variety of environments.

How do you respond to Searle's "Chinese room" argument? Searle's Chinese room argument attacks our functional position in a similar way by arguing that a system may appear to be intelligent without really understanding anything [Sea80]. From our perspective, whether or not an agent understands what it is doing is only important to the extent that it affects the measurable performance of the agent. If the performance is identical, as Searle seems to suggest, then whether or not the room with Searle inside understands the meaning of what is going on is of no practical concern; indeed it is not even clear to us how to define "understanding" if its presence has no measurable effects. So long as the system as a whole has the powerful properties required for universal machine intelligence, then we have the kind of extremely general and powerful machine that we desire. On the other hand, if "understanding" does have a measurable impact on an agent's performance in some well defined situations, then it is of interest to us. In which case, because Υ measures performance in all well defined situations, it follows that Υ is in part a measure of how much understanding an agent has.

But you don't deal with consciousness (or creativity, imagination, freewill, emotion, love, soul, etc.) We apply the same argument to consciousness, emotions, freewill, creativity, the soul and other such things. Our goal is to build powerful and flexible machines and thus these somewhat vague properties are only relevant to our goal to the extent to which they have some measurable effect on performance in some well defined environment. If no such measurable effect exists, then they are not relevant to our objective. Of course this is not the same as saying that these things do not exist. The question is whether they are relevant or not. We would consider creativity, appropriately defined, to have a significant impact on an agent's ability to adapt to challenging environments. Perhaps the same is also true of emotions, freewill and other qualities.

Universal intelligence is impossible due to the No-Free-Lunch theorem. Some, such as Edmonds [Edm06], argue that universal intelligence is impossible due to Wolpert's

so called “No Free Lunch” theorem [WM97]. However this theorem, or any of the standard variants on it, cannot be applied to universal intelligence for the simple reason that we have not taken a uniform distribution over the space of environments.

It is conceivable that there might exist some more general kind of “No Free Lunch” theorem for agents that limits their maximal intelligence according to our definition. Clearly any such result would have to apply only to computable agents given that the incomputable AIXI agent faces no such limit. If such a result were true, it would suggest that our definition of intelligence is perhaps too broad in its scope. Currently we know of no such result.

Interestingly, if it could be shown that an upper limit on Υ existed for feasible machines and that humans performed above this limit, then this would prove that humans have some incomputable element to their operation, perhaps consciousness, which is of real practical significance to their performance.

5.3 Conclusion

“... we need a definition of intelligence that is applicable to machines as well as humans or even dogs. Further, it would be helpful to have a relative measure of intelligence, that would enable us to judge one program more or less intelligent than another, rather than identify some absolute criterion. Then it will be possible to assess whether progress is being made ...”

— W. L. Johnson [Joh92]

Given the obvious significance of formal definitions of intelligence for research, and calls for more direct measures of machine intelligence to replace the problematic Turing test and other imitation based tests, little work has been done in this area. In this paper we have attempted to tackle this problem by taking an informal definition of intelligence modelled on expert definitions of human intelligence, and then generalise and formalise it. We believe that the resulting mathematical definition captures the concept of machine intelligence in a very powerful and yet elegant way. Furthermore, by considering alternative, more tractable measures of complexity, practical tests that estimate universal intelligence should be possible. Developing such tests will be the next major task in this direction of research.

The fact that we have stated our definition of machine intelligence in precise mathematical terms, rather than the more usual vaguely worded descriptions, means that there is no reason why criticisms of our approach should not be equally clear and precise. At the very least we hope that this in itself will help raise the debate over the definition and nature of machine intelligence to a new level of scientific rigour.

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